

Vellore Institute of Technology, Chennai
BECE407P - ASIC Design

Lab Report: 2R/1W CPU Register File

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Date: 27th October 2025

1.Aim:

To design, verify, and implement a complete RTL-to-GDSII flow for an 8-register, 32-bit CPU register file. The design must be in synthesizable Verilog and feature two read ports (dual-port read), one write port, a synchronous write operation, and an asynchronous active-low reset.

2.EDA Tools Used:

- **Simulation (Functional Verification):** Cadence Xcelium
 - **Code Coverage:** Cadence Nclaunch & Incisive Metric Center (IMC)
 - **Synthesis & DFT:** Cadence Genus
 - **Logic Equivalence Check (LEC):** Cadence Conformal
 - **ATPG:** Cadence Modus
 - **Physical Design (Place & Route):** Cadence Innovus
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3.Detailed description of the Designs

The design implements a 8x32-bit register file with two read ports and one write port.

- **Core Storage:** The core is a 2D array: reg [DATA_WIDTH-1:0] registers [0:REG_COUNT-1].
- **Write Logic:** A single always block sensitive to posedge clk or negedge rst_n handles all sequential logic.
 - **Asynchronous Reset:** The if (!rst_n) condition implements the active-low

asynchronous reset. A for loop iterates from 0 to 7 to set all 8 registers to 32'b0.

- **Synchronous Write:** The else if (we_w) condition implements the synchronous write. On a positive clock edge, if the write-enable signal (we_w) is high, the data on data_w is written to the register specified by addr_w.
- **Read Logic:** Two combinational assign statements implement the two read ports (data_a and data_b). The outputs reflect the data at registers[addr_a] and registers[addr_b] respectively. This creates an asynchronous-read, "read-before-write" memory, where reading and writing to the same address in the same cycle will result in the *old* data being read.

4.Procedure

The full RTL-to-GDSII flow was executed.

Step 4.1: Functional Verification (Xcelium)

The RTL was verified for logical correctness. The testbench (register_file_tb.v) was specifically designed to test the asynchronous reset, the reset's ability to override a write, and multiple read/write operations to different addresses to ensure functionality and improve coverage.

- **Command:** xrun register_file.v register_file_tb.v -access +rwc -sv
- **Result:** The simulation passed, with the testbench console printing TEST PASSED for all test cases, including the toggle coverage test.

Step 4.2: Code Coverage (Nclaunch & Xcelium IMC)

Code coverage was run to ensure the testbench provided adequate stimulus.

- **Command:** nclaunch (to open the GUI).
- **Result:** The initial testbench showed low coverage. An improved testbench (register_file_tb.v) was created to add tests for reset-override and to write various bit-toggling patterns (FFFFFFFF, 00000000, AAAAAAAAAA, 55555555) to all registers. This significantly improved the Toggle and Expression coverage.

Step 4.3: Synthesis & DFT (Genus)

The verified RTL was synthesized into a gate-level netlist with scan chains.

- **Command:** genus -f register_file_script_dft.tcl
- **Process:**
 1. The script read the 90nm slow.lib, the RTL (register_file.v), and the timing constraints (register_file_sdc.g).

2. DFT was configured for muxed_scan style with a SE (Scan Enable) port.
3. The design was synthesized (syn_generic, syn_map, syn_opt).
4. A single scan chain (top_chain) was defined and connected.
5. The final DFT-inserted netlist (register_file_generated_netlist_dft.v) and other reports were generated.
6. ATPG files were written to the test_scripts/ directory.

Step 4.4: Logic Equivalence Check (Conformal)

LEC was performed to verify the synthesized netlist against the original RTL.

- **Command:** lec -XI -nogui -color -64 -dofile register_file_lec.do
- **Process:** The register_file_lec.do script read the golden (RTL) and revised (netlist) designs. DFT pins (scan_in, scan_out, SE) were correctly ignored or constrained.
- **Result:** The comparison passed, reporting **Equivalent**.

Step 4.5: ATPG (Modus)

The ATPG files from Genus were used to generate test patterns.

- **Directory:** synthesis/test_scripts/
- **Command:** modus -f register_file_modus.tcl
- **Result:** Modus successfully processed the design and generated test vectors, confirming the scan chain was correctly inserted and the design is testable.

Step 4.6: Physical Design (Innovus)

The final netlist was implemented using the Place & Route flow.

- **Command:** innovus (GUI)
- **Process:**
 1. **Import:** The register_file_generated_netlist_dft.v and .sdc files were imported. MMMC was configured with slow.lib, fast.lib, GPDK 90nm LEF files, and the captable.
 2. **Floorplan:** A floorplan with 0.7 core utilization and an aspect ratio of 1 was defined.
 3. **Power:** Power rings and stripes were created for vdd and vss.
 4. **Placement:** Standard cells were placed.
 5. **CTS:** The clock tree was synthesized.
 6. **Routing:** NanoRoute was used for global and detailed routing.
 7. **Post-Route:** Filler cells were added to complete the design.
 8. **Export:** The final layout was saved as register_file.gds.

5.RTL Codes

Design: register_file.v

```
`timescale 1ns / 1ps

// 8-Register, 32-bit, 2-Read/1-Write Register File
module register_file #(
    parameter DATA_WIDTH = 32,
    parameter REG_COUNT = 8,
    parameter ADDR_WIDTH = 3 // 2^3 = 8 registers
) (
    input wire clk,
    input wire rst_n, // Asynchronous active-low reset

    // Port A (Read)
    input wire [ADDR_WIDTH-1:0] addr_a,
    output wire [DATA_WIDTH-1:0] data_a,

    // Port B (Read)
    input wire [ADDR_WIDTH-1:0] addr_b,
    output wire [DATA_WIDTH-1:0] data_b,

    // Port W (Write)
    input wire we_w, // Write Enable
    input wire [ADDR_WIDTH-1:0] addr_w,
    input wire [DATA_WIDTH-1:0] data_w
);

// Core register storage
reg [DATA_WIDTH-1:0] registers [0:REG_COUNT-1];

// Loop variable for reset
integer i;

// Write Logic: Synchronous write, Asynchronous reset
always @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        // Asynchronous reset: clear all registers
        for (i = 0; i < REG_COUNT; i = i + 1) begin
            registers[i] <= 32'b0;
        end
    end else if (we_w) begin
```



```

        // Synchronous write on posedge clk
        registers[addr_w] <= data_w;
    end
end

// Read Logic: Combinational (asynchronous)
// This creates a standard "read-before-write" file.
// The value read is the one from the previous clock cycle.
assign data_a = registers[addr_a];
assign data_b = registers[addr_b];

endmodule

```

Testbench: register_file_tb.v

```

`timescale 1ns / 1ps

module register_file_tb;

    // Parameters
    localparam DATA_WIDTH = 32;
    localparam ADDR_WIDTH = 3;
    localparam CLK_PERIOD = 10;

    // Testbench Signals
    reg clk;
    reg rst_n;
    reg [ADDR_WIDTH-1:0] addr_a;
    reg [ADDR_WIDTH-1:0] addr_b;
    reg we_w;
    reg [ADDR_WIDTH-1:0] addr_w;
    reg [DATA_WIDTH-1:0] data_w;
    wire [DATA_WIDTH-1:0] data_a;
    wire [DATA_WIDTH-1:0] data_b;

    // Instantiate the DUT
    register_file #(
        .DATA_WIDTH(DATA_WIDTH),
        .ADDR_WIDTH(ADDR_WIDTH)
    ) dut (
        .clk(clk),
        .rst_n(rst_n),

```

```

        .addr_a(addr_a),
        .data_a(data_a),
        .addr_b(addr_b),
        .data_b(data_b),
        .we_w(we_w),
        .addr_w(addr_w),
        .data_w(data_w)
    );

// Clock Generator
initial begin
    clk = 0;
    forever #(CLK_PERIOD / 2) clk = ~clk;
end

// Test Sequence
initial begin
    $dumpfile("wave.vcd");
    $dumpvars(0, dut);

    // 1. Initialize and Reset
    rst_n = 0; // Assert active-low reset
    we_w = 0;
    addr_a = 1;
    addr_b = 2;
    addr_w = 0;
    data_w = 32'hAAAAAAAA;

    #(CLK_PERIOD * 1.5); // Wait for reset to propagate

    // --- NEW TEST 1: Check Reset Override ---
    $display("\nT=%0t: NEW TEST 1: Check Reset Override", $time);
    we_w = 1; // Try to write during reset
    addr_w = 4;
    data_w = 32'hFFFFFFF;

    #(CLK_PERIOD * 1.0);
    rst_n = 1; // De-assert reset

    @(negedge clk);
    $display("T=%0t: Reset released.", $time);

    // 2. Check reset (and override)

```

```

@(posedge clk);
we_w = 0;
addr_a = 1; // Should be 0
addr_b = 4; // Should be 0 (write was ignored)

@(posedge clk);
if (data_a === 32'b0 && data_b === 32'b0) begin
    $display("T=%0t: TEST PASSED - Registers reset to 0 and reset override successful.",
$time);
end else begin
    $display("T=%0t: TEST FAILED - Reset failed. data_a=%h, data_b=%h", $time, data_a,
data_b);
end

// 3. Write to R3
we_w = 1;
addr_w = 3;
data_w = 32'hDEADBEEF;
$display("\nT=%0t: Writing 0xDEADBEEF to R3.", $time);

@(posedge clk);

// 4. Write to R5, Read from R3 (check prev write) and R1 (check reset)
we_w = 1;
addr_w = 5;
data_w = 32'h12345678;
addr_a = 3; // Read R3
addr_b = 1; // Read R1
$display("T=%0t: Writing 0x12345678 to R5. Reading R3 and R1.", $time);

@(posedge clk);
if (data_a === 32'hDEADBEEF && data_b === 32'b0) begin
    $display("T=%0t: TEST PASSED - R3=0xDEADBEEF, R1=0x0.", $time);
end else begin
    $display("T=%0t: TEST FAILED - R3 read failed. data_a=%h, data_b=%h", $time,
data_a, data_b);
end

// 5. Read R5 and R3 (check both writes)
we_w = 0;
addr_a = 5; // Read R5
addr_b = 3; // Read R3
$display("\nT=%0t: Reading R5 and R3.", $time);

```

```

    @(posedge clk);
    if (data_a === 32'h12345678 && data_b === 32'hDEADBEEF) begin
        $display("T=%0t: TEST PASSED - R5=0x12345678, R3=0xDEADBEEF.", $time);
    end else begin
        $display("T=%0t: TEST FAILED - R5/R3 read failed. data_a=%h, data_b=%h", $time,
data_a, data_b);
    end

    // --- NEW TEST 2: Full Toggle Coverage ---
    $display("\nT=%0t: NEW TEST 2: Full Toggle Coverage", $time);

    // Write all 1s and 0s
    we_w = 1;
    addr_w = 0; data_w = 32'hFFFFFFFF; @(posedge clk);
    addr_w = 6; data_w = 32'h00000000; @(posedge clk);

    // Write checkerboard patterns
    addr_w = 2; data_w = 32'hAAAAAAAA; @(posedge clk);
    addr_w = 4; data_w = 32'h55555555; @(posedge clk);

    // Write to last register
    addr_w = 7; data_w = 32'hCAFECAFE; @(posedge clk);
    we_w = 0;

    // Read back all new values
    $display("T=%0t: Reading all written values for toggle test.", $time);
    addr_a = 0; addr_b = 6; @(posedge clk);
    if (data_a !== 32'hFFFFFFFF || data_b !== 32'h00000000)
        $display("TEST FAILED: R0/R6 read incorrect.");

    addr_a = 2; addr_b = 4; @(posedge clk);
    if (data_a !== 32'hAAAAAAAA || data_b !== 32'h55555555)
        $display("TEST FAILED: R2/R4 read incorrect.");

    addr_a = 7; addr_b = 7; @(posedge clk);
    if (data_a !== 32'hCAFECAFE)
        $display("TEST FAILED: R7 read incorrect.");

    $display("\nT=%0t: Toggle test complete.", $time);
    $finish;
end

```

```
endmodule
```

SDC: register_file_sdc.g

```
create_clock -name clk -period 10 -waveform {0 5} [get_ports "clk"]
set_clock_transition -rise 0.1 [get_clocks "clk"]
set_clock_transition -fall 0.1 [get_clocks "clk"]
set_clock_uncertainty 0.01 [get_ports "clk"]
```

```
# Constrain all inputs relative to the clock
```

```
set_input_delay -max 1.0 [get_ports "rst_n"] -clock [get_clocks "clk"]
set_input_delay -max 1.0 [get_ports "we_w"] -clock [get_clocks "clk"]
set_input_delay -max 1.0 [get_ports "addr_a"] -clock [get_clocks "clk"]
set_input_delay -max 1.0 [get_ports "addr_b"] -clock [get_clocks "clk"]
set_input_delay -max 1.0 [get_ports "addr_w"] -clock [get_clocks "clk"]
set_input_delay -max 1.0 [get_ports "data_w"] -clock [get_clocks "clk"]
```

```
# Constrain all outputs relative to the clock
```

```
set_output_delay -max 1.0 [get_ports "data_a"] -clock [get_clocks "clk"]
set_output_delay -max 1.0 [get_ports "data_b"] -clock [get_clocks "clk"]
```

Genus Script: register_file_script_dft.tcl

```
set_db init_lib_search_path /home/install/FOUNDRY/digital/90nm/dig/lib
set_db init_hdl_search_path /home/student/Desktop/22BEC1204_ASIC_DESIGN/work
read_libs slow.lib
read_hdl register_file.v
elaborate
read_sdc register_file_sdc.g
```

```
set_db dft_scan_style muxed_scan
set_db dft_prefix dft_
define_shift_enable -name SE -active high -create_port SE
check_dft_rules
```

```
set_db syn_generic_effort medium
set_db syn_map_effort medium
set_db syn_opt_effort medium
syn_generic
syn_map
```

syn_opt

check_dft_rules

set_db design:register_file .dft_min_number_of_scan_chains 1

define_scan_chain -name top_chain -sdi scan_in -sdo scan_out -create_ports

connect_scan_chains -auto_create_chains

syn_opt -incr

report_scan_chains

write_dft_atpg -library /home/install/FOUNDRY/digital/90nm/dig/lib

write_hdl > register_file_generated_netlist_dft.v

write_sdf -timescale ns -nonegchecks -recrem split -edges check_edge -setuphold split >

register_file_generated_delays_dft.sdf

write_sdc > register_file_generated_sdc_dft.sdc

write_scandef > register_file_generated_scanDEF.scandef

gui_show

LEC Script: register_file_lec.do

set log file register_file_generated_LEC.log -replace

read library /home/student/22BEC1204_ASIC_DESIGN/LEC/slowlib_verilog_generated.v

-verilog -both

read design /home/student/22BEC1204_ASIC_DESIGN/work/register_file.v -verilog -golden

read design

/home/student/22BEC1204_ASIC_DESIGN/synthesis/register_file_generated_netlist_dft.v

-verilog -revised

add ignored inputs scan_in -revised

add ignored outputs scan_out -revised

add pin constraints 0 SE -revised

set system mode lec

add compare point -all

compare

report verification

set gui on

Modus Script: register_file_modus.tcl

build_model -workdir /home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts

-designsource register_file.test_netlist.v -techlib

Applications Places IMC (64) [Analysis]

en Nov 4 17:58

IMC (64) [Analysis - Metrics]

File View Analysis Help

Load Date Context Views Analyze Refinement Report

Relative Elements

Verification Hierarchy

Name	Overall Average Grade	Overall Covered	Assertion Status Grade
Recursive	(no filter)	(no filter)	(no filter)
Types	83.49%	488 / 548 (89.05%)	n/a
register_file_tb	83.49%	244 / 274 (89.05%)	n/a
register_file	70.21%	137 / 160 (85.62%)	n/a
Instances	96.76%	107 / 114 (93.86%)	n/a
register_file_tb	83.49%	244 / 274 (89.05%)	n/a
dut	96.76%	107 / 114 (93.86%)	n/a

Showing 7 items

Metrics Source Attributes

Name	Overall Average Grade	Overall Covered
Overall	96.76%	107 / 114 (93.86%)
Code	96.76%	107 / 114 (93.86%)
Block	100%	6 / 6 (100%)
Expression	n/a	0 / 0 (n/a)
Toggle	93.52%	101 / 108 (93.52%)
FSM	n/a	0 / 0 (n/a)
Functional	n/a	0 / 0 (n/a)
Assertion	n/a	0 / 0 (n/a)
CoverGroup	n/a	0 / 0 (n/a)

Showing 9 items

Loaded Run: /cdev/workspaces/est

work [Lab_4_LEC_guide.tx... register_file_tb.v (- student@ic312work NCLaunch /home/st... Design Browser 1 - S... Console - SimVison Waveform 1 - SimVis... IMC (64) [Analysis - ...]

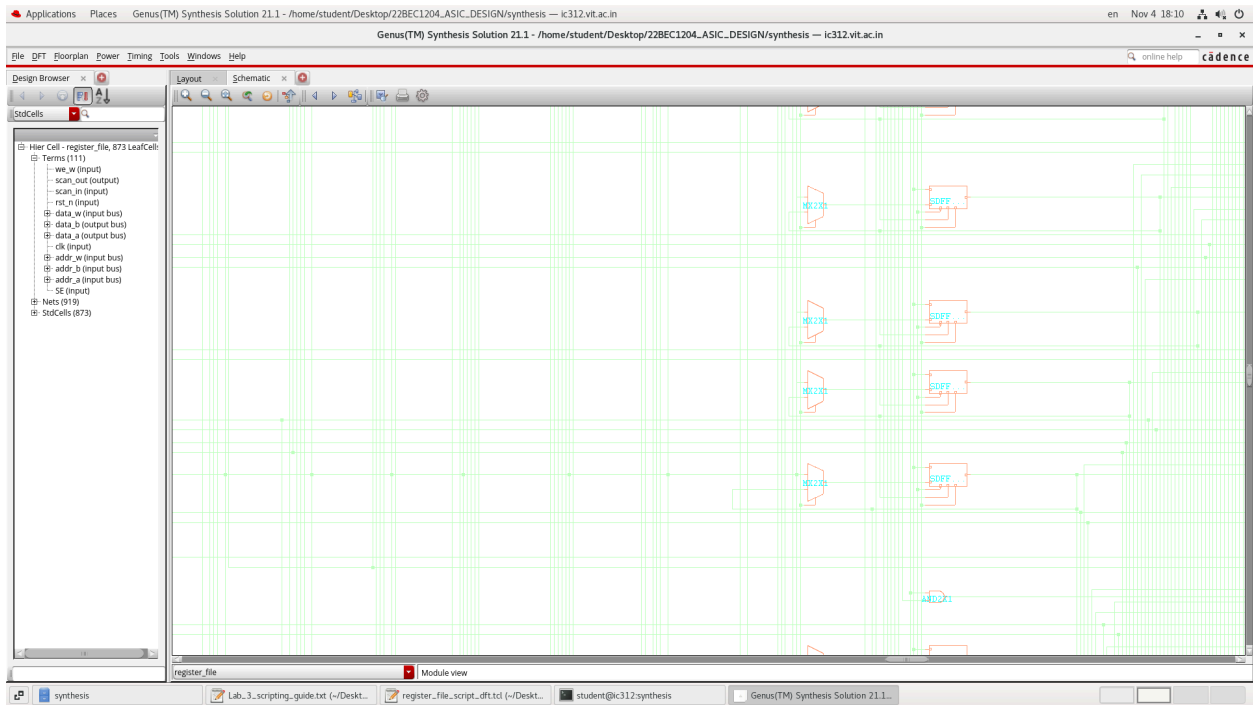
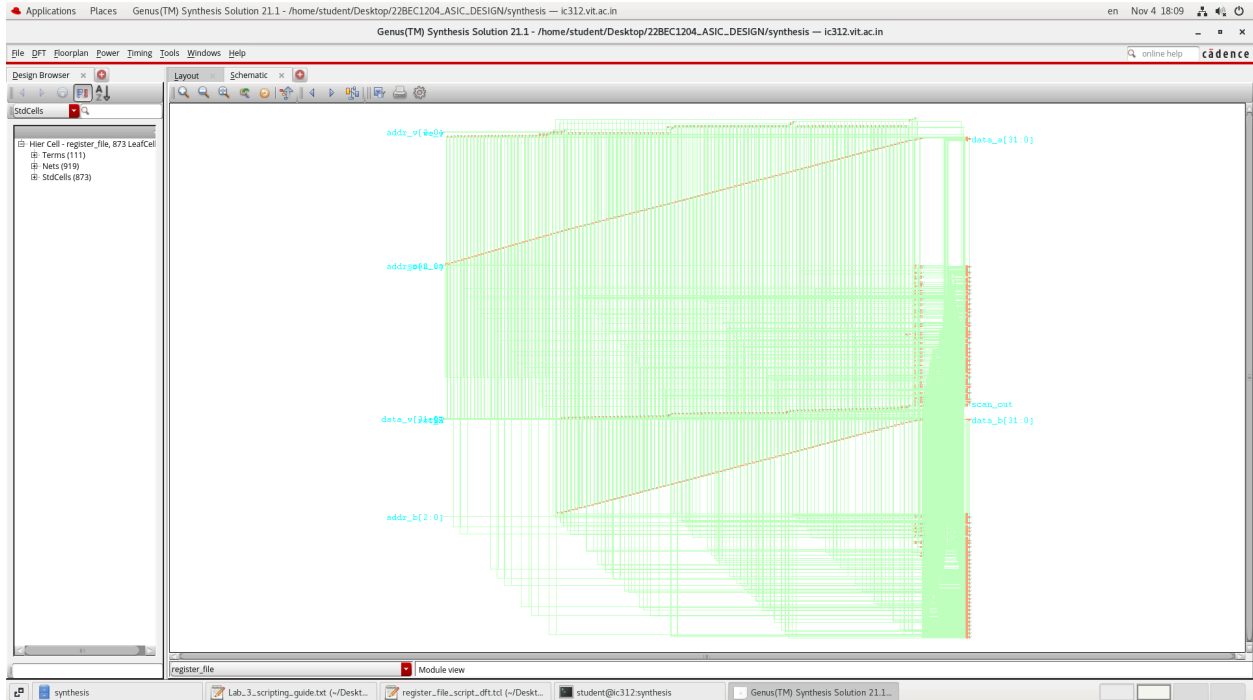
Applications Places Terminal

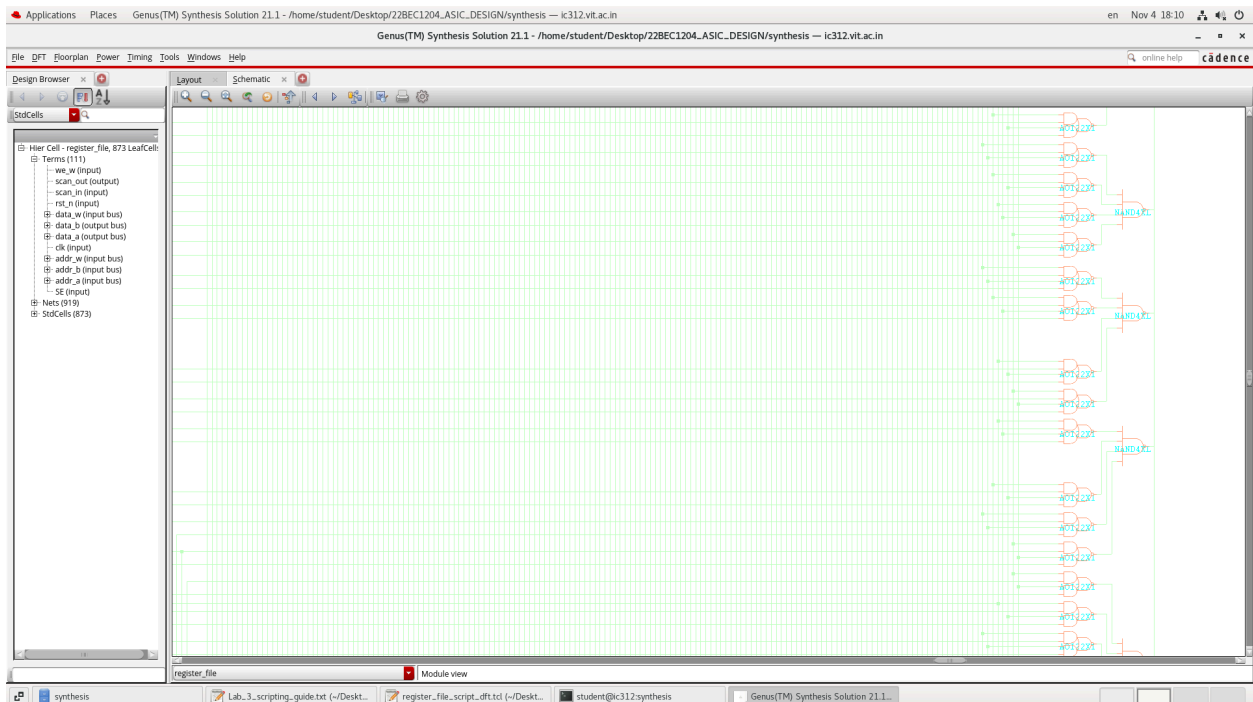
en Nov 4 18:05

student@ic312:work

```
File Edit View Search Terminal Help
[student@ic312 work]$ xrun register_file.v register_file_tb.v -access +rwc -sv -gui &
[1] 36224
[student@ic312 work]$ TOOL: xrun 22.09-s003: Started on Nov 04, 2025 at 17:46:12 IST
TOOL: xrun(64) 22.09-s003: Started on Nov 04, 2025 at 17:46:12 IST
xrun(64): 22.09-s003: (c) Copyright 1995-2022 Cadence Design Systems, Inc.
file: register_file.v
module worklib.register_file.v
errors: 0, warnings: 0
file: register_file_tb.v
module worklib.register_file_tb.v
errors: 0, warnings: 0
Caching library 'worklib' ..... Done
Elaborating the design hierarchy:
Top level design units:
register_file_tb
xmelab: *W,05EMEL: This SystemVerilog design will be simulated as per IEEE 1800-2009 SystemVerilog simulation semantics. Use -disable_sem2009 option for turning off SV 2009 simulation semantics.
Building instance overlay tables: ..... Done
Generating native compiled code:
worklib.register_file.v <0x2f5cfa69>
streams: 5, words: 2190
worklib.register_file_tb.v <0x6aaa7cca>
streams: 9, words: 17996
Building instance specific data structures.
Loading native compiled code: ..... Done
Design hierarchy summary:
Instances Unique
Modules: 2 2
Registers: 9 9
Scalar wires: 3 -
Vectorized wires: 0 -
Always blocks: 1 1
Initial blocks: 2 2
Cont. assignments: 2 2
Pseudo assignments: 7 7
Simulation timescale: 1ps
Writing initial simulation snapshot: worklib.register_file_tb.v
xmsim: *W,NOMTDGUI: Multi-Threaded Dumping is disabled for interactive debug mode.
xmsim: *W,05EM2009: This SystemVerilog design is simulated as per IEEE 1800-2009 SystemVerilog simulation semantics. Use -disable_sem2009 option for turning off SV 2009 simulation semantics.
Relinquished control to SimVision...
xcellum> xrun(64) 22.09-s003: Exiting on Nov 04, 2025 at 17:46:26 IST (total: 00:00:14)
[1] Done xrun register_file.v register_file_tb.v -access +rwc -sv -gui
[student@ic312 work]$
[student@ic312 work]$ nclaunch
nclaunch(64): 15.20-s086: (c) Copyright 1995-2020 Cadence Design Systems, Inc.
[student@ic312 work]$
[student@ic312 work]$ nclaunch
nclaunch(64): 15.20-s086: (c) Copyright 1995-2020 Cadence Design Systems, Inc.
[student@ic312 work]$
```

synthesis [Lab_3_scripting_guide.txt (+/Deskt... register_file_script.dftd (+/Deskt... student@ic312:work]





Verification Report		
Category		Count
1. Non-standard modeling options used:		0
2. Incomplete verification:		1
Command "add ignore outputs" used:	yes	
3. User modification to design:		0
4. Conformal Constraint Designer clock domain crossing checks recommended:		0
5. Design ambiguity:		0
6. Compare Results:		PASS
=====		
// Command: set gui on		
// Switching to GUI mode		
█		



Google Gemini - Google Chrome



LEC



register_file_lec.do (~)

```
Applications Places Terminal Nov 5 5:32 PM student@ic312:test_scripts

File Edit View Search Terminal Help

1.2.1.4.8 | 74.49 | 0.51 | 74.49 | 21.55 | 0.00 | 21.55 | 258 | 257 | 7224 |
1.2.1.4.9 | 76.06 | 1.57 | 76.06 | 21.55 | 0.00 | 21.55 | 258 | 257 | 7482 |
1.2.1.4.10 | 86.95 | 10.89 | 86.95 | 21.55 | 0.00 | 21.55 | 258 | 257 | 7740 |
1.2.1.4.11 | 87.43 | 0.48 | 87.43 | 21.55 | 0.00 | 21.55 | 258 | 257 | 7998 |
1.2.1.4.12 | 93.66 | 6.23 | 93.66 | 21.55 | 0.00 | 21.55 | 258 | 257 | 8256 |
1.2.1.4.13 | 94.15 | 0.49 | 94.15 | 21.55 | 0.00 | 21.55 | 258 | 257 | 8514 |
1.2.1.4.14 | 95.20 | 1.85 | 95.20 | 21.55 | 0.00 | 21.55 | 258 | 257 | 8772 |
1.2.1.4.15 | 95.72 | 0.52 | 95.72 | 21.55 | 0.00 | 21.55 | 258 | 257 | 9030 |
1.2.1.4.16 | 97.83 | 2.11 | 97.83 | 21.55 | 0.00 | 21.55 | 258 | 257 | 9288 |
1.2.1.5.1 | 88.99 | 1.16 | 88.99 | 21.55 | 0.00 | 21.55 | 258 | 257 | 9546 |
1.2.1.5.2 | 99.53 | 0.54 | 99.53 | 21.55 | 0.00 | 21.55 | 258 | 257 | 9804 |
1.2.1.5.3 | 99.96 | 0.43 | 99.96 | 21.55 | 0.00 | 21.55 | 258 | 257 | 10062 |
1.2.1.5.4 | 99.97 | 0.01 | 99.97 | 21.55 | 0.00 | 21.55 | 258 | 257 | 10320 |
1.2.1.5.5 | 99.98 | 0.01 | 99.98 | 21.55 | 0.00 | 21.55 | 258 | 257 | 10578 |
1.2.1.5.6 | 100.00 | 0.02 | 100.00 | 21.55 | 0.00 | 21.55 | 259 | 0 | 10837 |
[end TVE_050]

INFO (TDA-001): Maximum Memory used during the run and Cumulative Time in hours:minutes:seconds:

Total Memory = 8,851,824 bytes

CPU Time = 0:00:00.00
Elapsed Time = 0:00:00.25 [end TDA_001]

Date Ended: Wednesday Nov 05 17:31:56 2025 IST

INFO (TVE-002): Verilog write vectors has completed. [end TVE_002]

-----
* Message Summary
*
Count Number First Instance of Message Text
-----
INFO Messages...
1 INFO (TVE-001): Verilog write vectors started.
1 INFO (TVE-002): Verilog write vectors has completed.
4 INFO (TVE-003): Verilog write vectors output file will be: /home/student/22BEC1204 ASIC_DESIGN/synthesis/test_scripts/testresults/verilog/register_file_modus_generated_results.cylemap.
2 INFO (TVE-004): Reading test section 1.1. Test section type equals scan.
2 INFO (TVE-005): Created 516 total cycles, of which 4 are test cycles, 512 are scan cycles, 0 are dynamic timed cycles and 0 are dynamic cycles that are not timed.
2 INFO (TVE-008): Created 256 total measures, of which 0 are PO measures and 256 are SO (Scan Out) measures.
1 INFO (TVE-050): TEST SEQUENCE COVERAGE ESTIMATE REPORT
3 INFO (TVE-103): There was no specified TEST offset for this clock 'pin clk'. A default clock offset of 8.000000 ns will be used.

For a detailed explanation of a message and a suggested user response execute 'msgHelp <message id>'. For example: msgHelp TDA-009

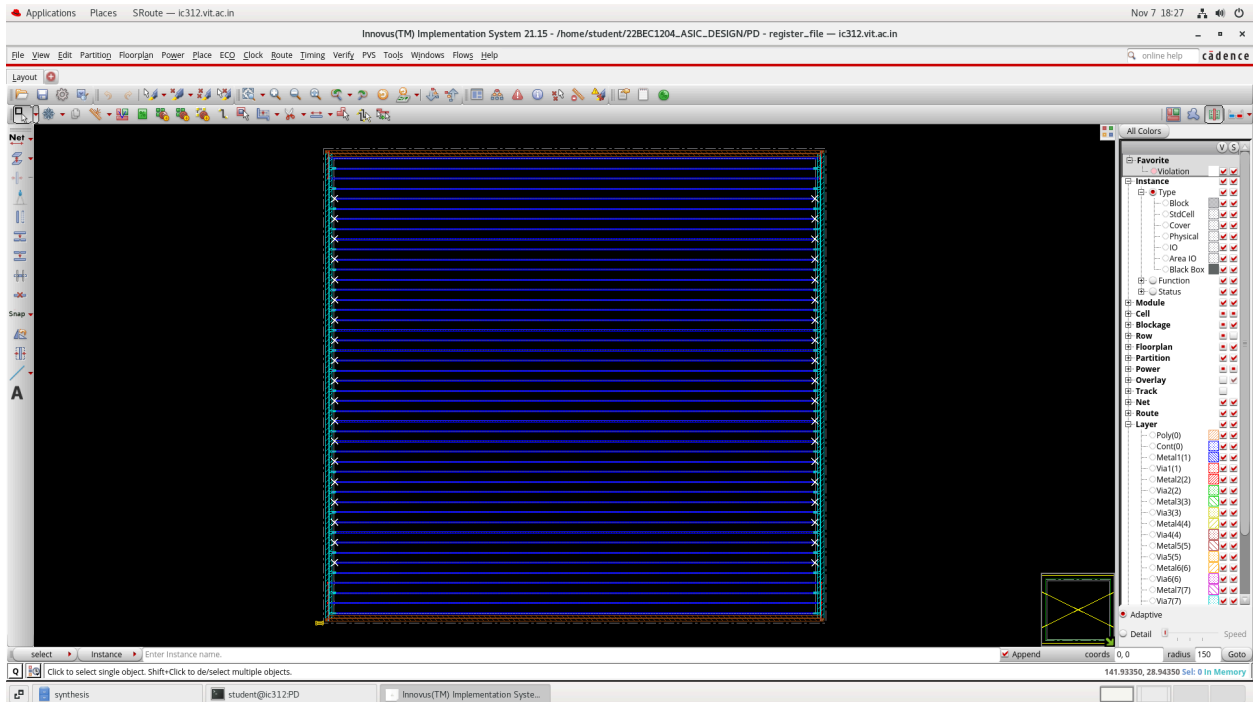
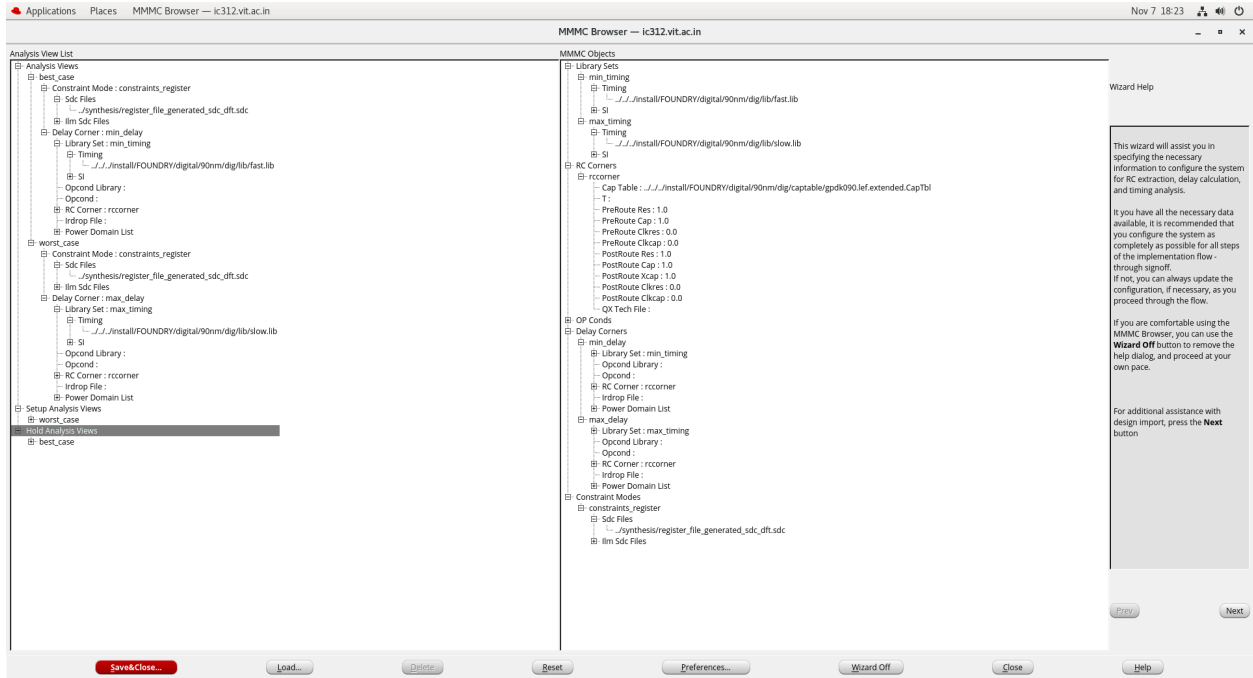
-----
#@ End verbose source register_file_modus.tcl
0
@modus:root:/ 2> █
```

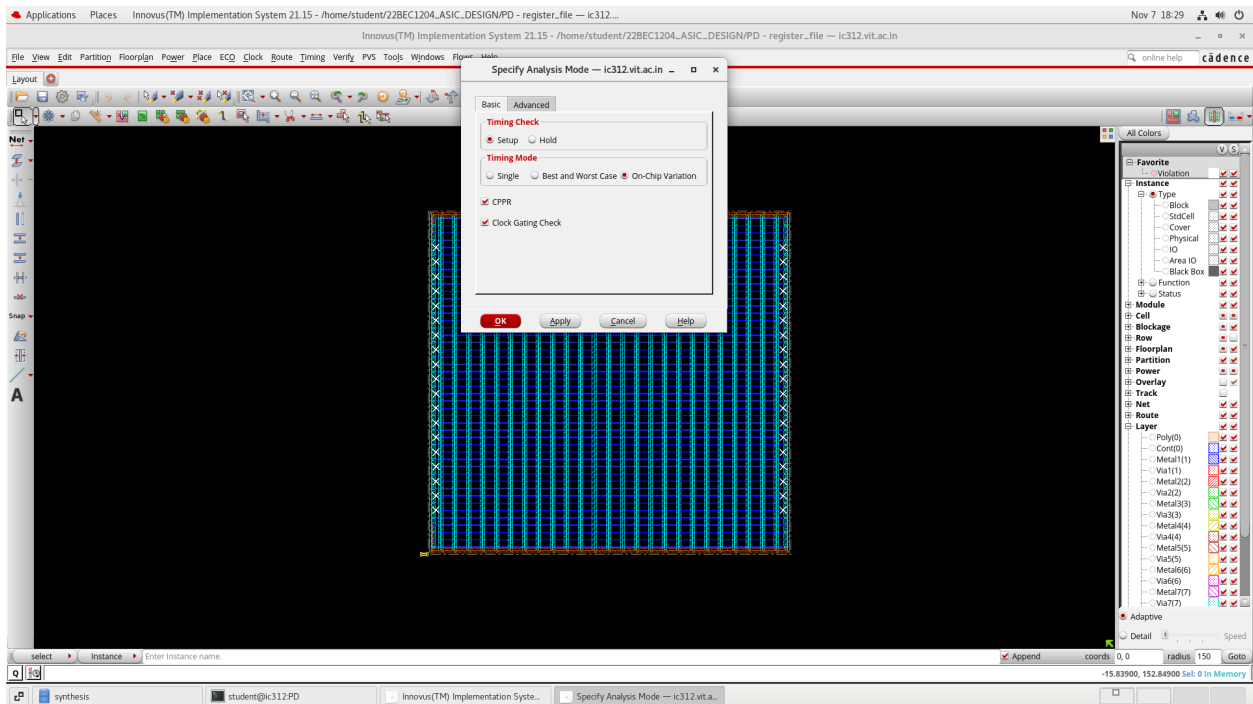
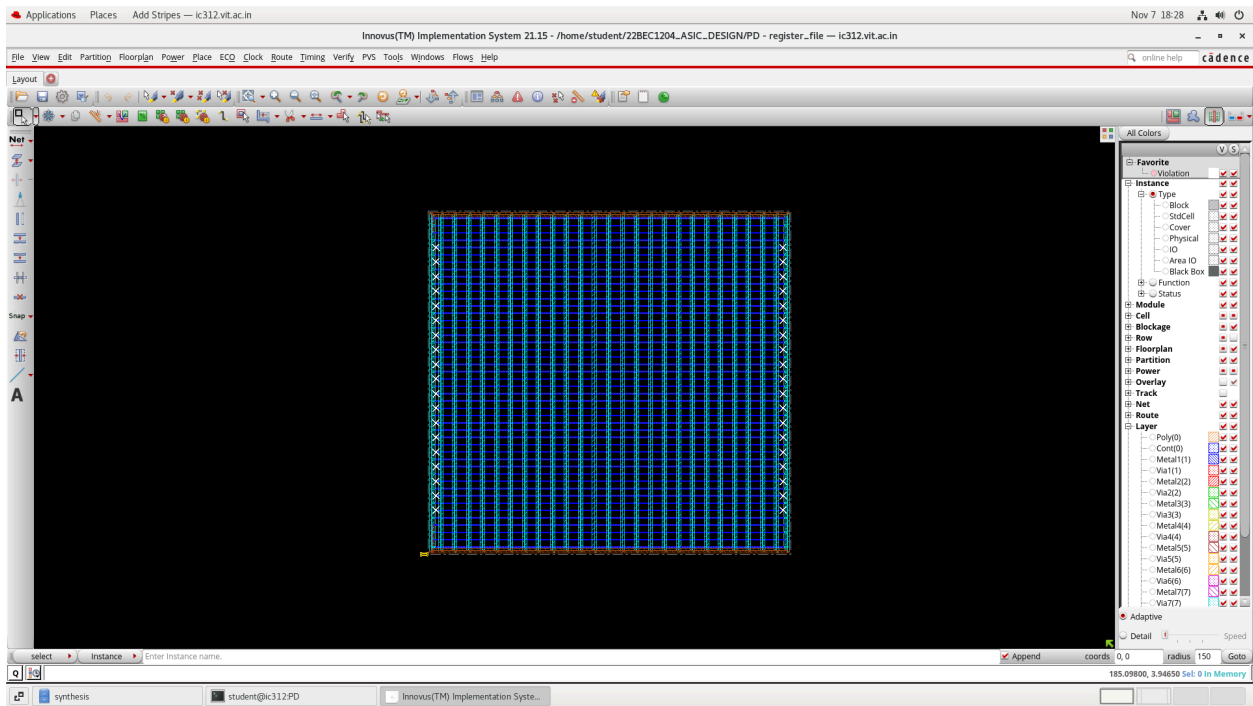
```
verilog student@ic312:test_scripts pipeline_control_modus_generated...

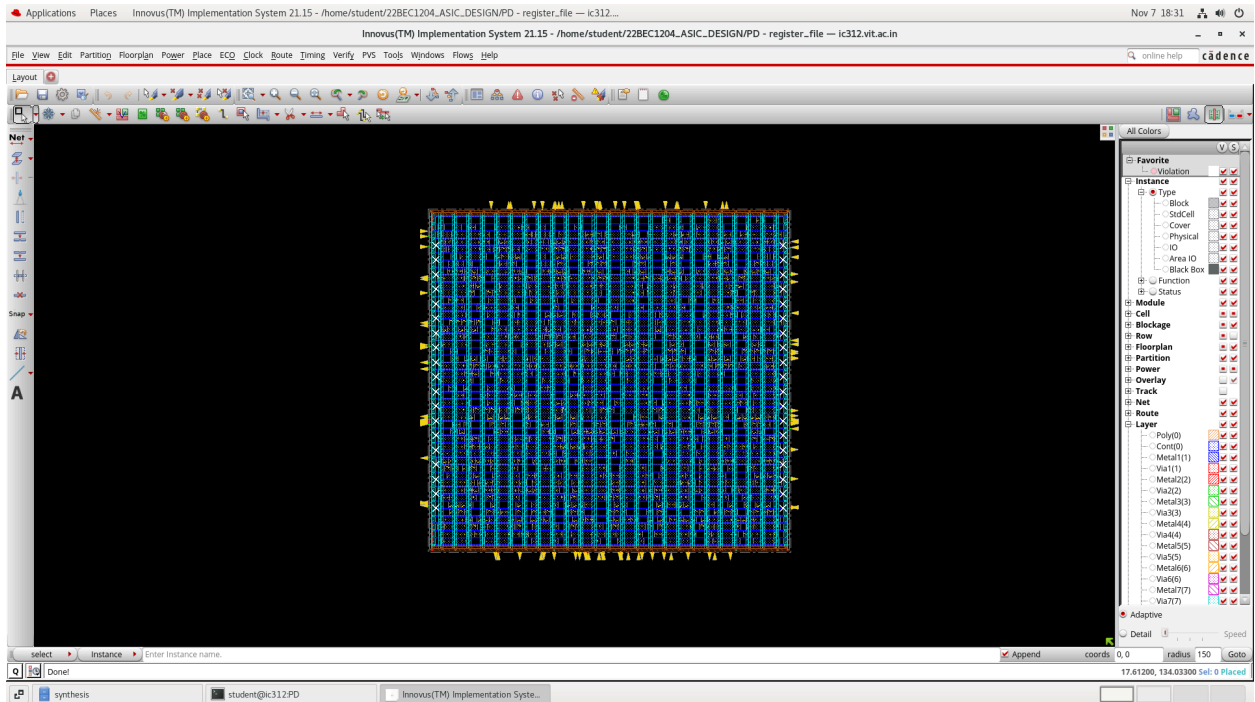
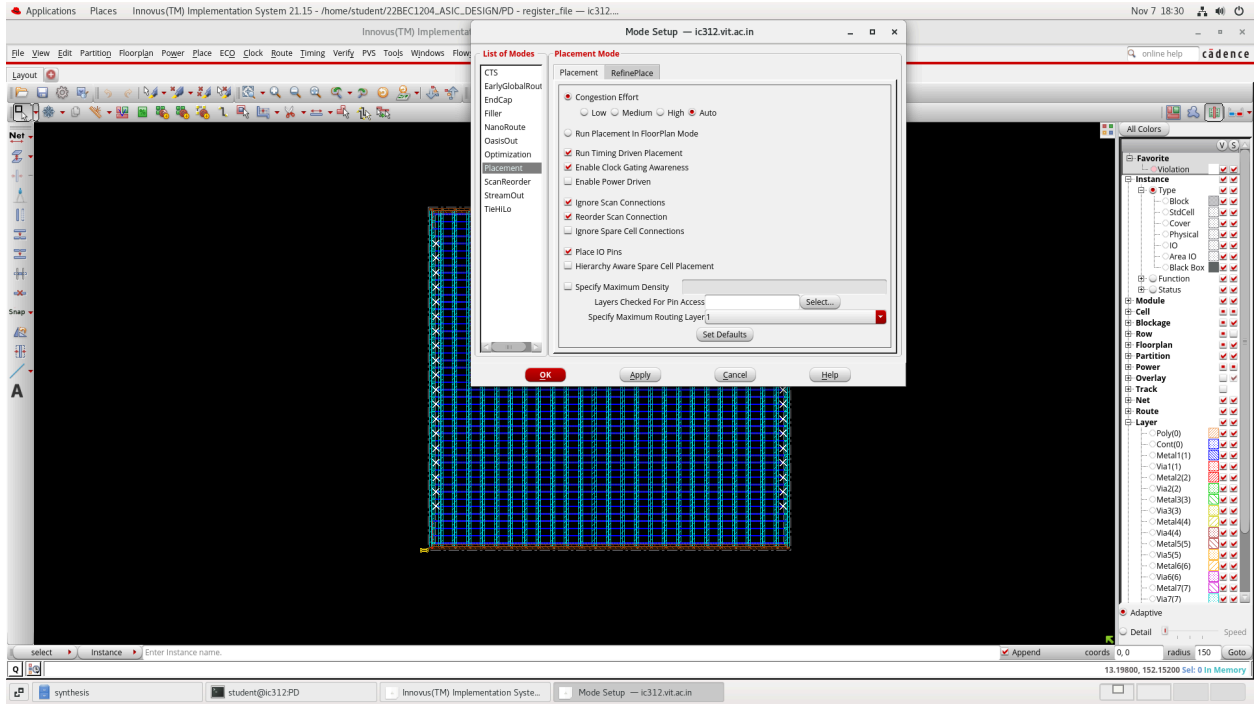
-----
* Message Summary
*
Count Number First Instance of Message Text
-----
INFO Messages...
1 INFO (TVE-001): Verilog write vectors started.
1 INFO (TVE-002): Verilog write vectors has completed.
4 INFO (TVE-003): Verilog write vectors output file will be: /home/student/22BEC1204 ASIC_DESIGN/synthesis/test_scripts/testresults/verilog/register_file_modus_generated_results.cylemap.
2 INFO (TVE-004): Reading test section 1.1. Test section type equals scan.
2 INFO (TVE-005): Created 516 total cycles, of which 4 are test cycles, 512 are scan cycles, 0 are dynamic timed cycles and 0 are dynamic cycles that are not timed.
2 INFO (TVE-008): Created 256 total measures, of which 0 are PO measures and 256 are SO (Scan Out) measures.
1 INFO (TVE-050): TEST SEQUENCE COVERAGE ESTIMATE REPORT
3 INFO (TVE-103): There was no specified TEST offset for this clock 'pin clk'. A default clock offset of 8.000000 ns will be used.

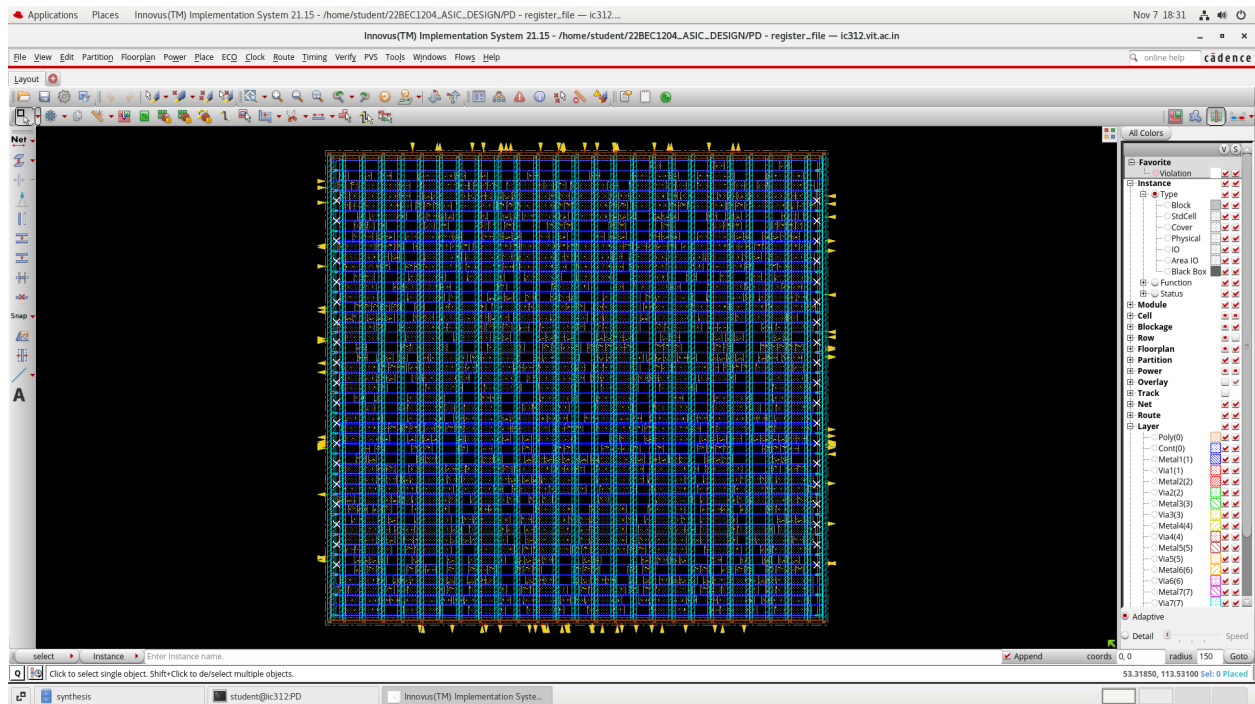
For a detailed explanation of a message and a suggested user response execute 'msgHelp <message id>'. For example: msgHelp TDA-009

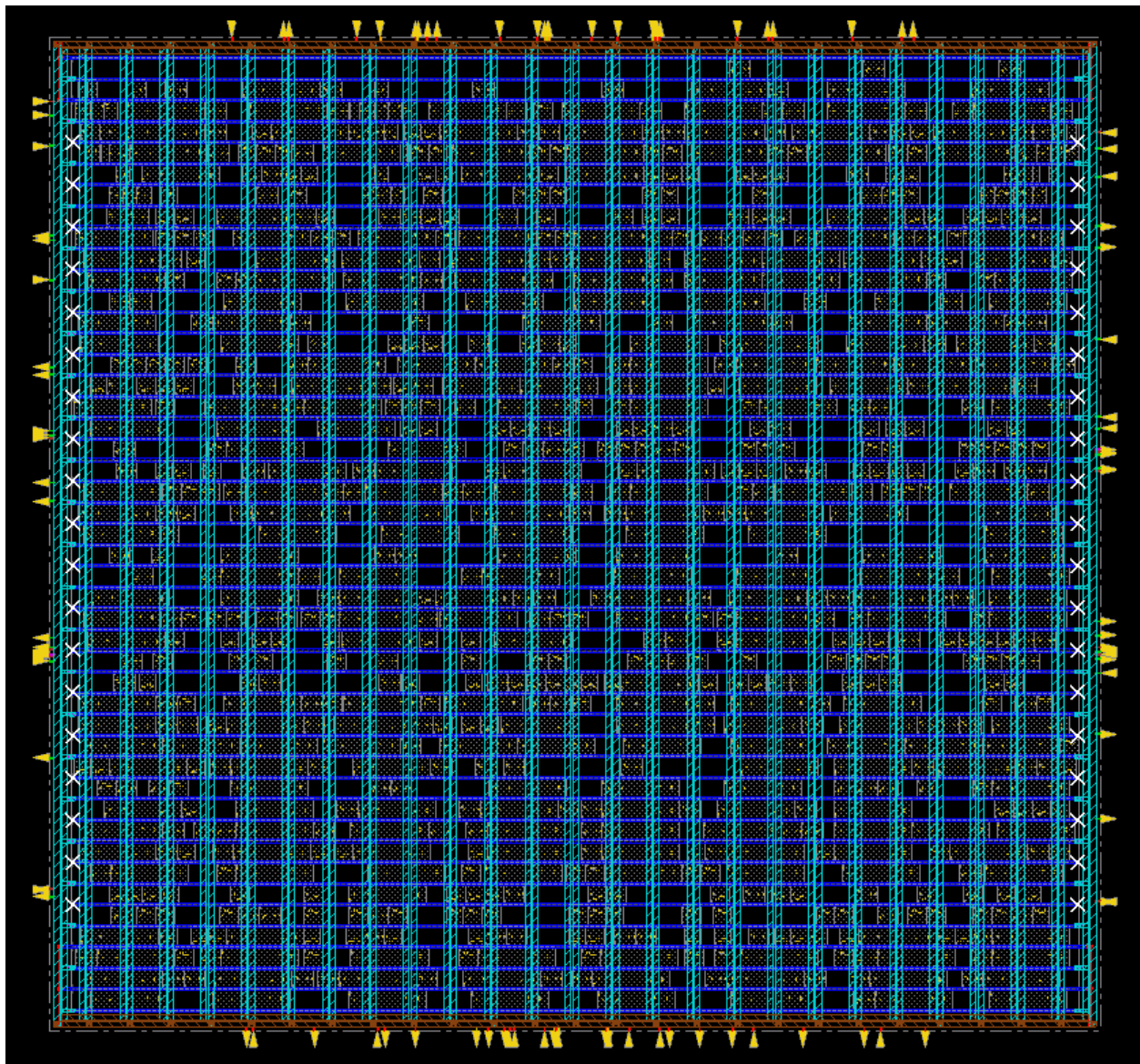
-----
#@ End verbose source register_file_modus.tcl
0
@modus:root:/ 2> █
```

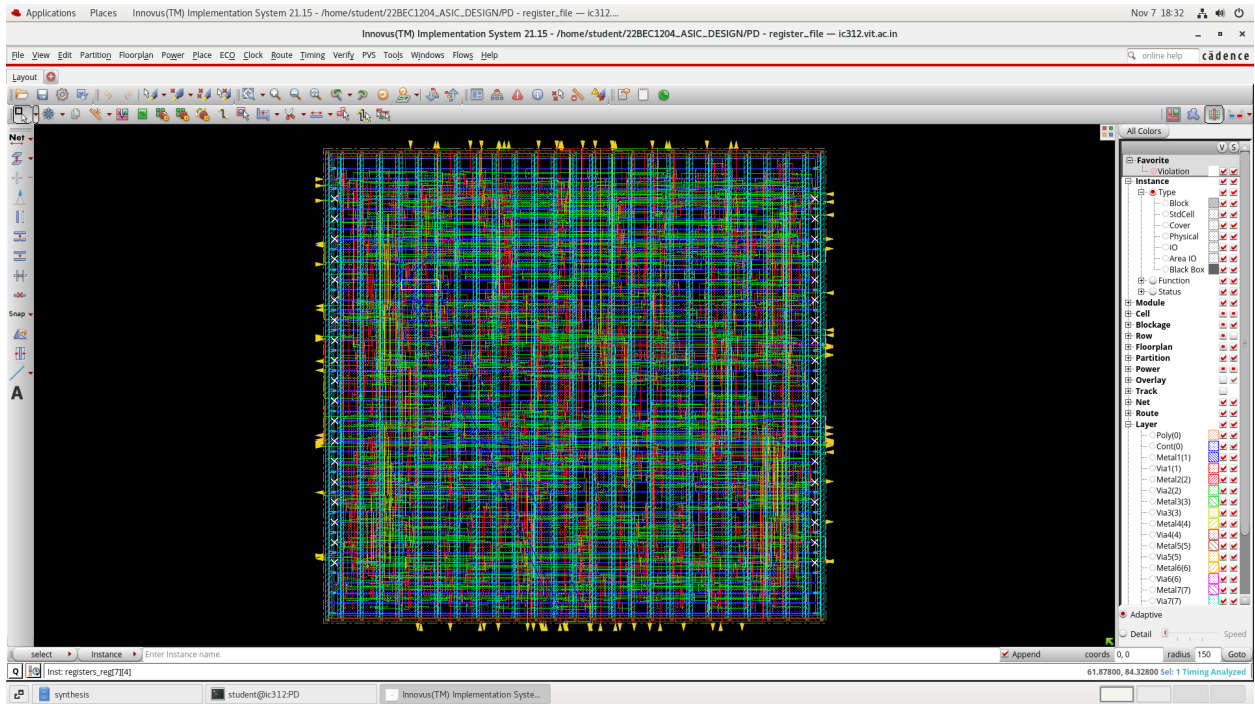
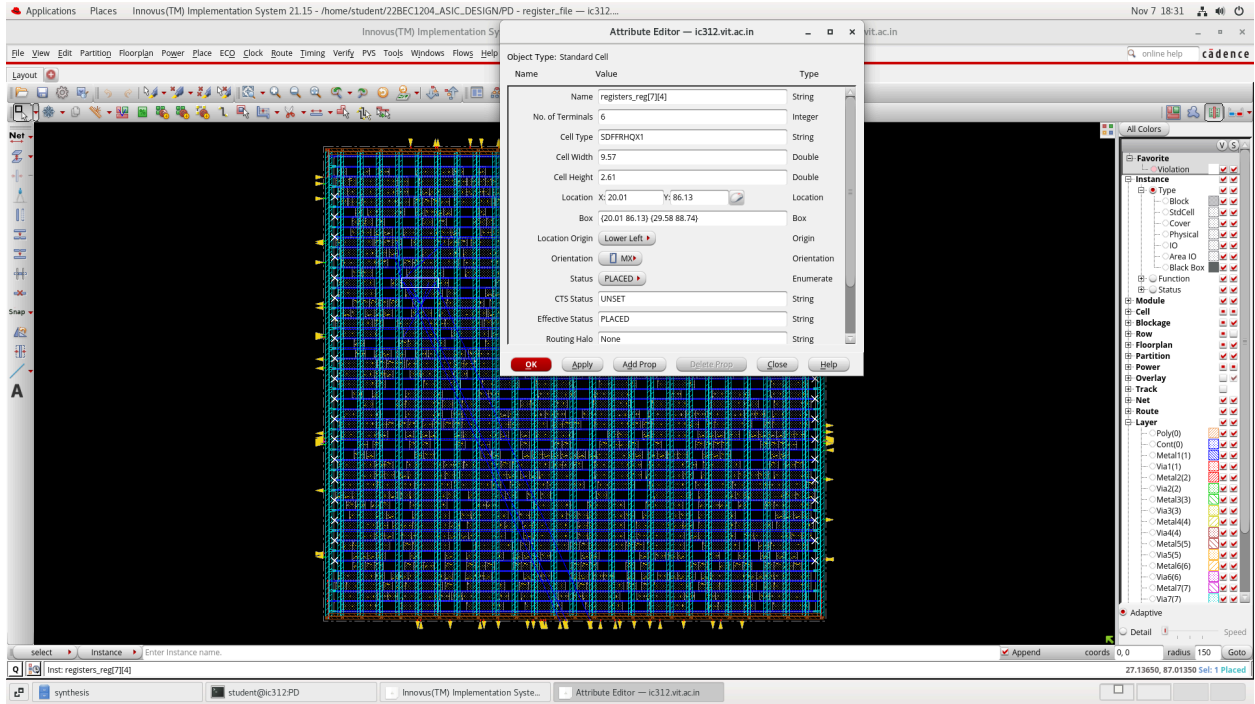



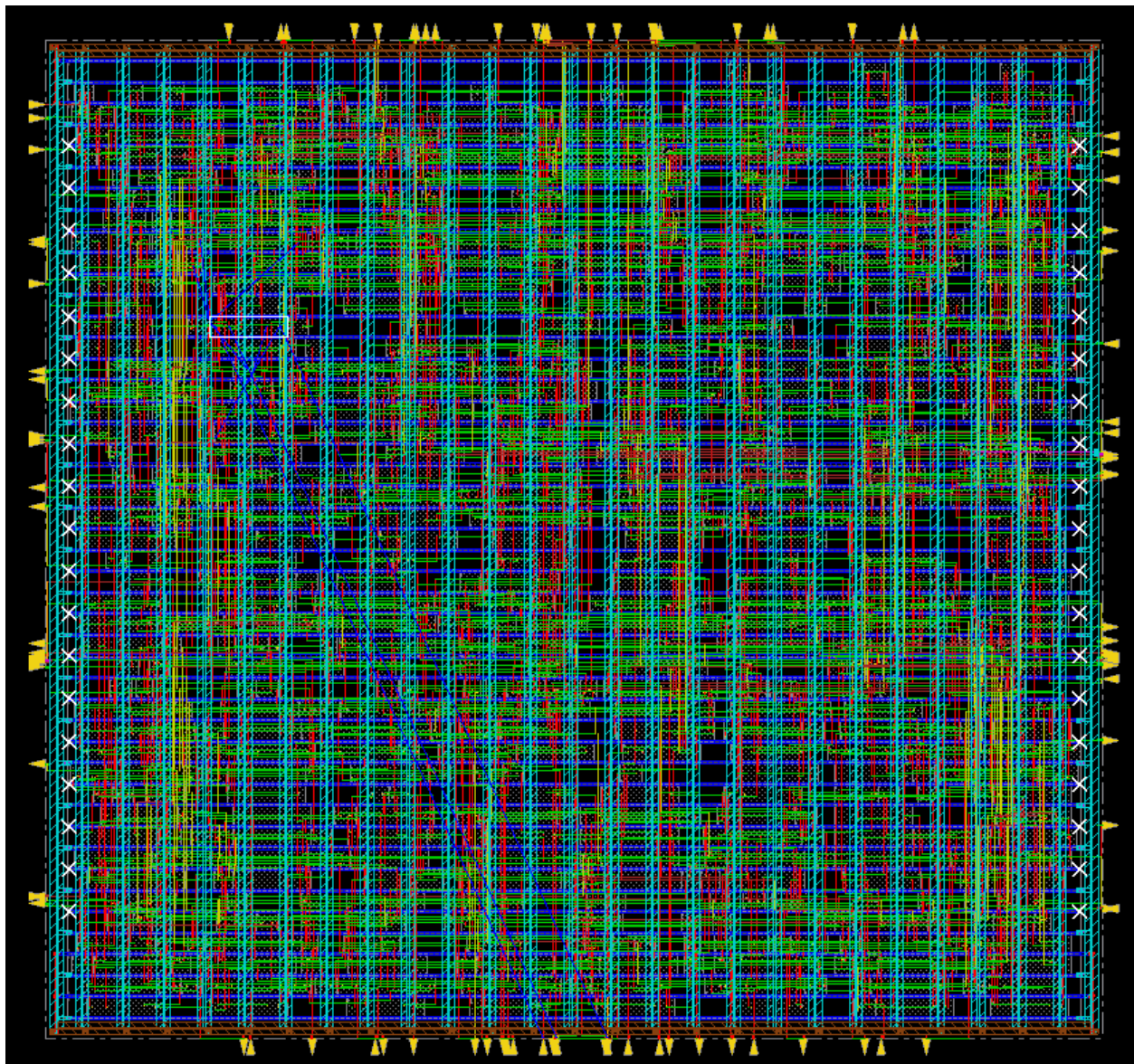


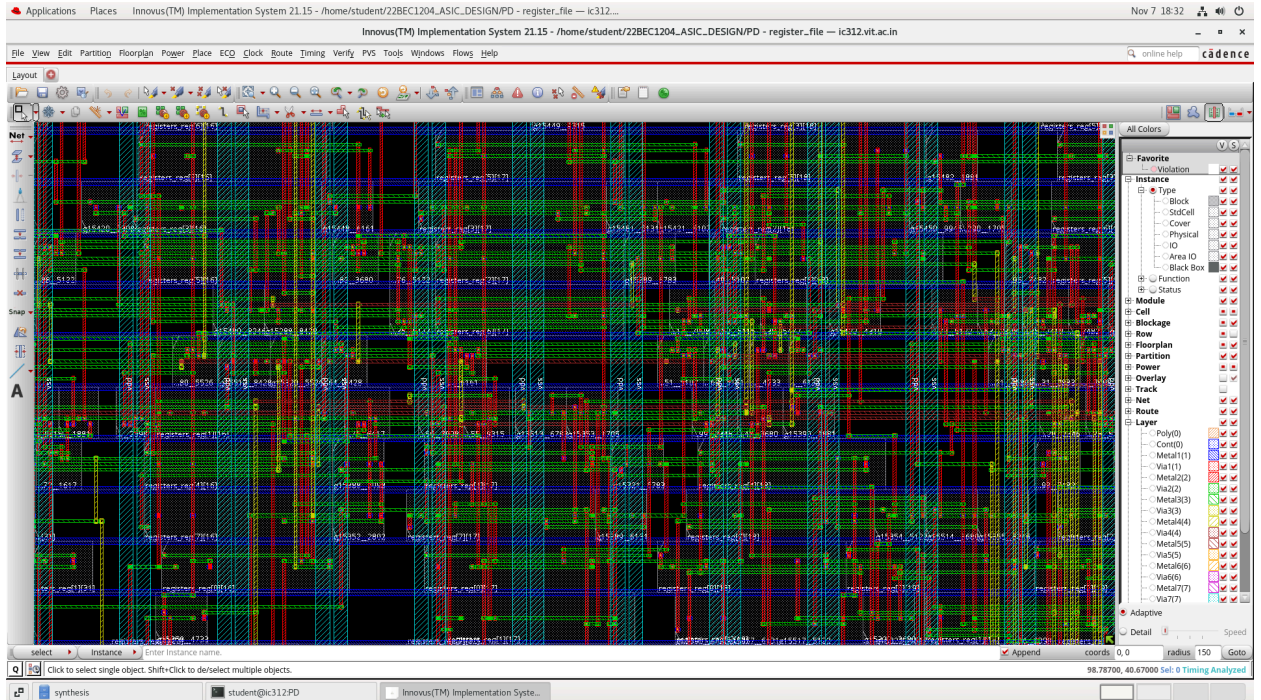


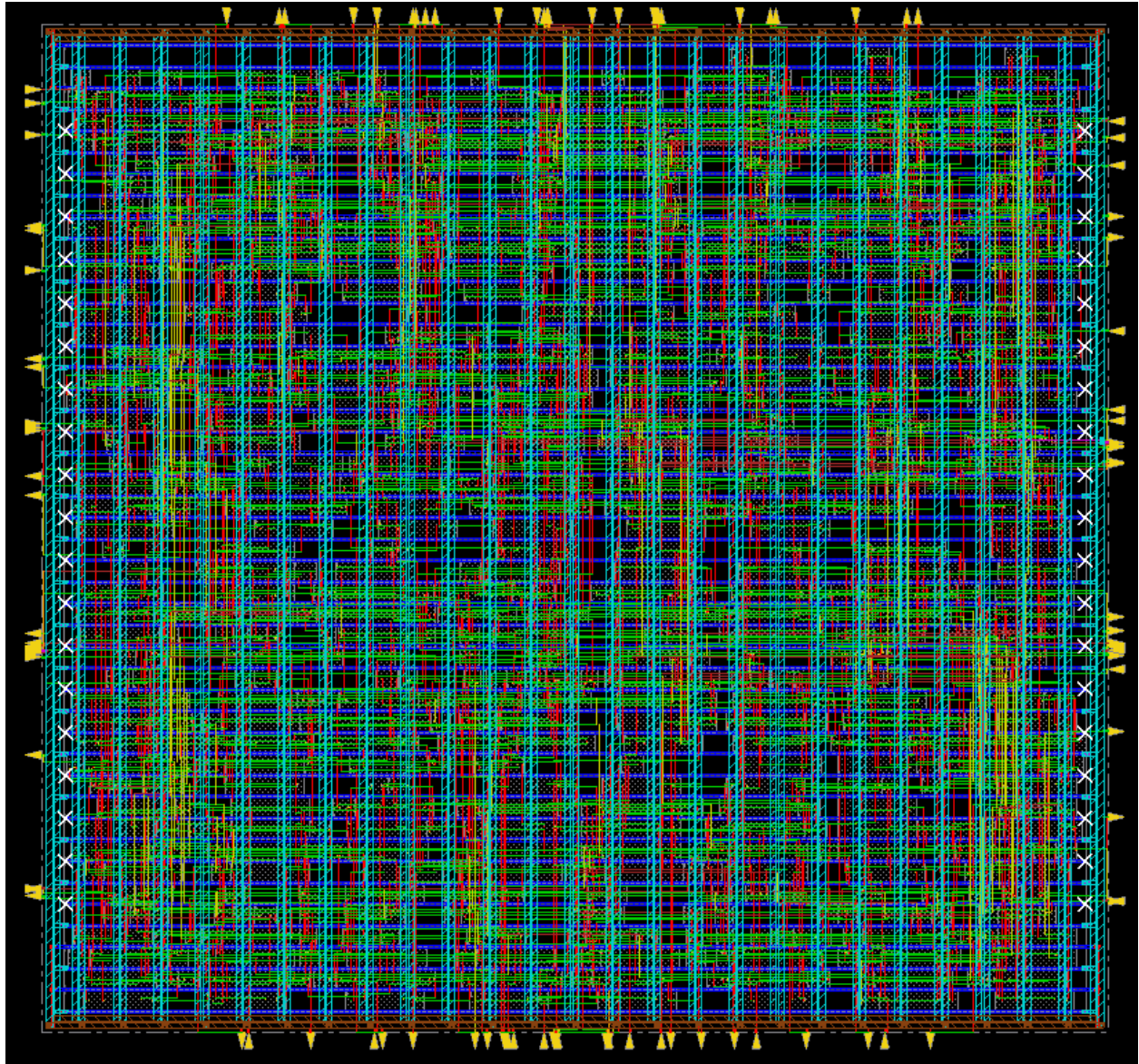


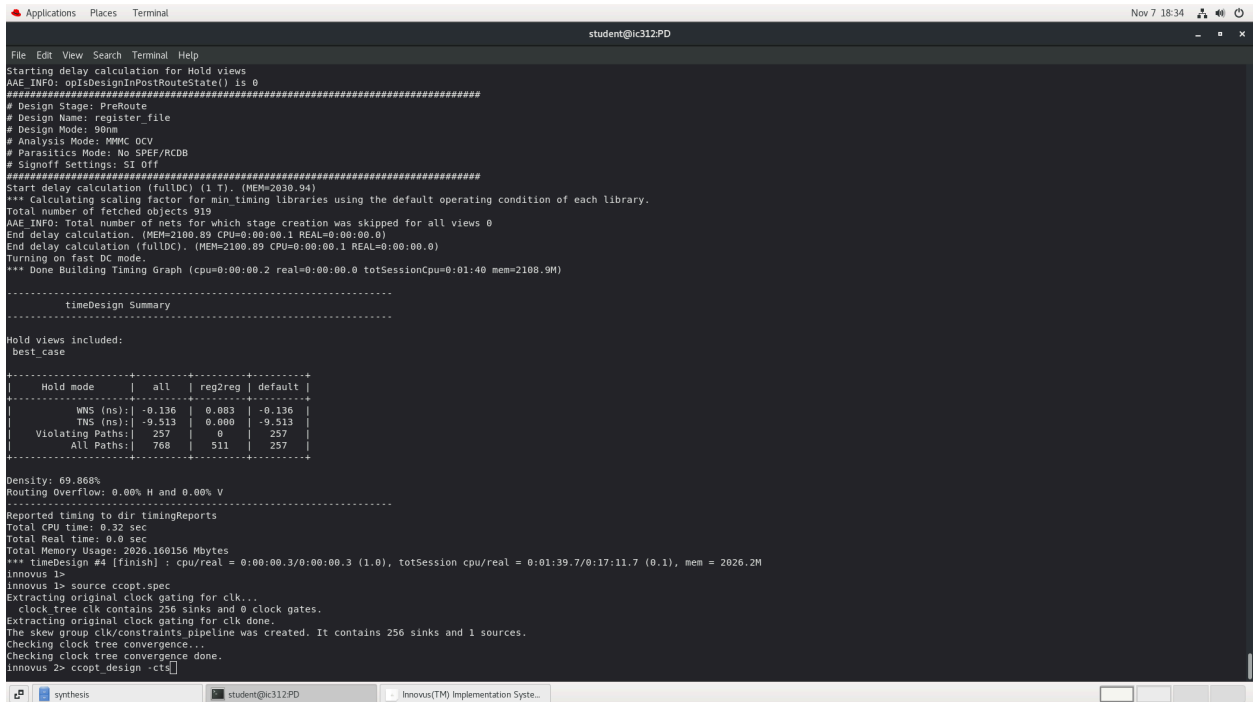
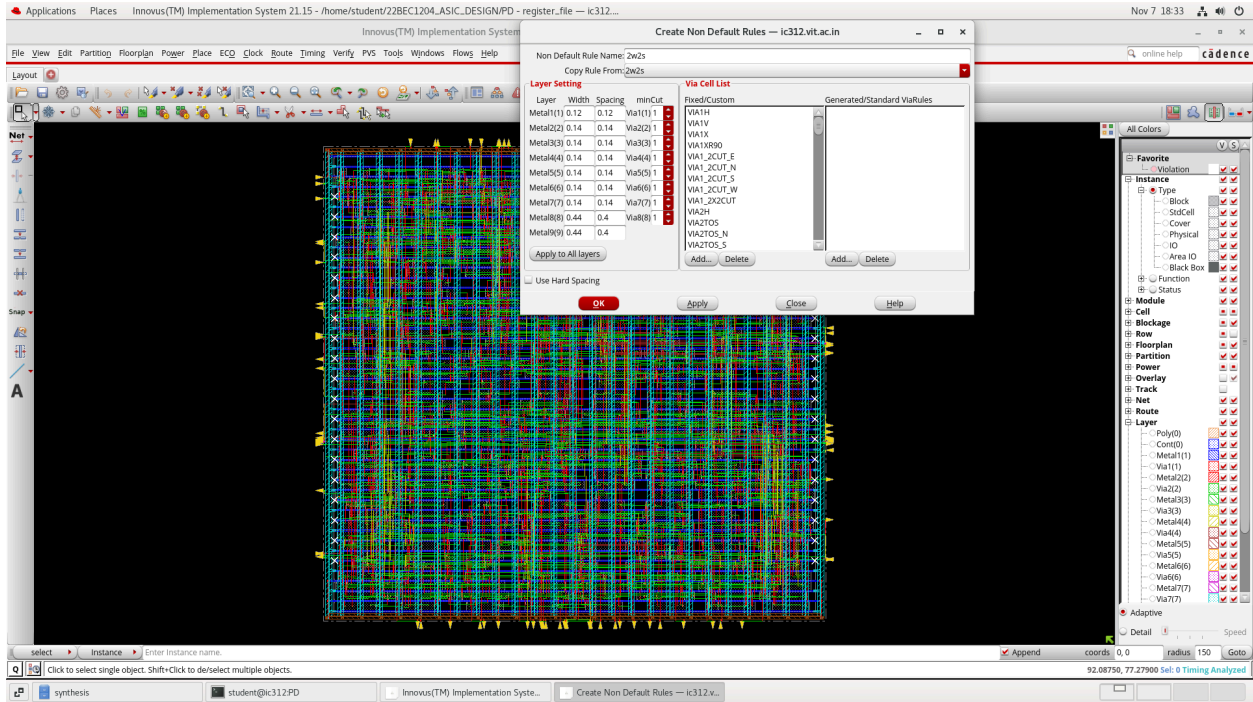












```
Applications Places Terminal Nov 7 18:34 student@ic312PD

File Edit View Search Terminal Help
No ideal or dont_touch nets found in the clock tree
No dont_touch hnets found in the clock tree
No dont_touch hpins found in the clock network.
Checking for illegal sizes of clock logic instances...
Checking for illegal sizes of clock logic instances done. (took cpu=0:00:00.0 real=0:00:00.0)
Validating CTS configuration done. (took cpu=0:00:00.0 real=0:00:00.0)

CCOpt configuration status: cannot run ccopt_design.
Check the log for details of problem(s) found:

-----
Design configuration problems
-----
One or more clock trees have configuration problems
-----

Clock tree configuration problems:
-----
Clock tree Problem
-----
clk Could not determine drivers to use
-----

CCOpt::Phase::PreparingToBalance done. (took cpu=0:00:00.1 real=0:00:00.1)
Runtime done. (took cpu=0:00:00.1 real=0:00:00.1)
Runtime Summary
=====
Clock Runtime: (53%) Core CTS 0.05 (Init 0.05, Construction 0.00, Implementation 0.00, eGRPC 0.00, PostConditioning 0.00, Other 0.00)
Clock Runtime: (0%) CTS services 0.00 (RefinePlace 0.00, EarlyGlobalClock 0.00, NanoRoute 0.00, ExtractRC 0.00, TimingAnalysis 0.00)
Clock Runtime: (46%) Other CTS 0.05 (Init 0.05, CongRepair/EGR-DP 0.00, TimingUpdate 0.00, Other 0.00)
Clock Runtime: (100%) Total 0.10

Leaving CCOpt scope - Cleaning up placement interface...
Leaving CCOpt scope - Cleaning up placement interface done. (took cpu=0:00:00.0 real=0:00:00.0)
**ERROR: 3
*** CTS #1 [finish] (ccopt_design #1) : cpu/real = 0:00:00.1/0:00:00.1 (1.0), totSession cpu/real = 0:01:48.2/0:19:13.9 (0.1), mem = 2034.5M
**ERROR: (IMPCCOPT-2196): Cannot run ccopt_design because the command prerequisites were not met. Review the previous error messages for more details about the failure.

*** Summary of all messages that are not suppressed in this session:
Severity ID Count Summary
ERROR IMPCCOPT-1135 1 CTS found neither inverters nor buffers ...
ERROR IMPCCOPT-2196 1 Cannot run ccopt_design because the comm...
WARNING IMPCCOPT-1183 1 The library has no usable balanced %ss f...
WARNING IMPCCOPT-1184 1 The library has no usable balanced %ss f...
*** Message Summary: 2 warning(s), 2 error(s)

*** ccopt_design #1 [finish] : cpu/real = 0:00:00.1/0:00:00.1 (0.9), totSession cpu/real = 0:01:48.2/0:19:13.9 (0.1), mem = 2034.5M
#% End ccopt_design (date=11/07 18:34:26, total cpu=0:00:00.1, real=0:00:01.0, peak res=1900.6M, current mem=1900.6M)

innovus > 
```

```
Applications Places Terminal Nov 7 18:35 student@ic312PD

File Edit View Search Terminal Help
No ideal or dont_touch nets found in the clock tree
No dont_touch hnets found in the clock tree
No dont_touch hpins found in the clock network.
Checking for illegal sizes of clock logic instances...
Checking for illegal sizes of clock logic instances done. (took cpu=0:00:00.0 real=0:00:00.0)
Validating CTS configuration done. (took cpu=0:00:00.0 real=0:00:00.0)

CCOpt configuration status: cannot run ccopt_design.
Check the log for details of problem(s) found:

-----
Design configuration problems
-----
One or more clock trees have configuration problems
-----

Clock tree configuration problems:
-----
Clock tree Problem
-----
clk Could not determine drivers to use
-----

CCOpt::Phase::PreparingToBalance done. (took cpu=0:00:00.1 real=0:00:00.1)
Runtime done. (took cpu=0:00:00.1 real=0:00:00.1)
Runtime Summary
=====
Clock Runtime: (53%) Core CTS 0.05 (Init 0.05, Construction 0.00, Implementation 0.00, eGRPC 0.00, PostConditioning 0.00, Other 0.00)
Clock Runtime: (0%) CTS services 0.00 (RefinePlace 0.00, EarlyGlobalClock 0.00, NanoRoute 0.00, ExtractRC 0.00, TimingAnalysis 0.00)
Clock Runtime: (46%) Other CTS 0.05 (Init 0.05, CongRepair/EGR-DP 0.00, TimingUpdate 0.00, Other 0.00)
Clock Runtime: (100%) Total 0.10

Leaving CCOpt scope - Cleaning up placement interface...
Leaving CCOpt scope - Cleaning up placement interface done. (took cpu=0:00:00.0 real=0:00:00.0)
**ERROR: 3
*** CTS #1 [finish] (ccopt_design #1) : cpu/real = 0:00:00.1/0:00:00.1 (1.0), totSession cpu/real = 0:01:48.2/0:19:13.9 (0.1), mem = 2034.5M
**ERROR: (IMPCCOPT-2196): Cannot run ccopt_design because the command prerequisites were not met. Review the previous error messages for more details about the failure.

*** Summary of all messages that are not suppressed in this session:
Severity ID Count Summary
ERROR IMPCCOPT-1135 1 CTS found neither inverters nor buffers ...
ERROR IMPCCOPT-2196 1 Cannot run ccopt_design because the comm...
WARNING IMPCCOPT-1183 1 The library has no usable balanced %ss f...
WARNING IMPCCOPT-1184 1 The library has no usable balanced %ss f...
*** Message Summary: 2 warning(s), 2 error(s)

*** ccopt_design #1 [finish] : cpu/real = 0:00:00.1/0:00:00.1 (0.9), totSession cpu/real = 0:01:48.2/0:19:13.9 (0.1), mem = 2034.5M
#% End ccopt_design (date=11/07 18:34:26, total cpu=0:00:00.1, real=0:00:01.0, peak res=1900.6M, current mem=1900.6M)

innovus > saveDesign DBS/cts.enc4
```



```
Applications Places Terminal Nov 7 18:35 student@ic312PD

File Edit View Search Terminal Help

*** Summary of all messages that are not suppressed in this session:
Severity ID Count Summary
ERROR IMPCCOPT-1135 1 CTS found neither inverters nor buffers ...
ERROR IMPCCOPT-2196 1 Cannot run ccopt design because the comm...
WARNING IMPCCOPT-1183 1 The library has no usable balanced %ss f...
WARNING IMPCCOPT-1184 1 The library has no usable balanced %ss f...
*** Message Summary: 2 warning(s), 2 error(s)

*** ccopt design #1 [finish] : cpu/real = 0:00:00.1/0:00:00.1 (0.9), totSession cpu/real = 0:01:48.2/0:19:13.9 (0.1), mem = 2834.5M
% End ccopt design (date=11/07 18:34:26, total cpu=0:00:00.1, real=0:00:01.0, peak res=1908.6M, current mem=1900.6M)

Innovus 3> saveDesign DBS/cts.enc4
% Begin save design ... (date=11/07 18:35:22, mem=1905.6M)
% Begin Save ccopt configuration ... (date=11/07 18:35:22, mem=1905.6M)
% End Save ccopt configuration ... (date=11/07 18:35:22, total cpu=0:00:00.0, real=0:00:00.0, peak res=1906.0M, current mem=1906.0M)
% Begin Save netlist data ... (date=11/07 18:35:22, mem=1906.0M)
Writing Binary db to DBS/cts.enc4.dat/register file.vbin in single-threaded mode...
% End Save netlist data ... (date=11/07 18:35:22, total cpu=0:00:00.0, real=0:00:00.0, peak res=1906.5M, current mem=1906.5M)
Saving symbol-table file ...
Saving congestion map file DBS/cts.enc4.dat/register file.route.congmap.gz ...
% Begin Save AAE data ... (date=11/07 18:35:23, mem=1906.5M)
Saving AAE Data ...
% End Save AAE data ... (date=11/07 18:35:23, total cpu=0:00:00.0, real=0:00:00.0, peak res=1906.5M, current mem=1906.5M)
Saving preference file DBS/cts.enc4.dat/gui.pref.tcl ...
Saving mode setting ...
Saving global file ...
% Begin Save floorplan data ... (date=11/07 18:35:23, mem=1908.4M)
Saving floorplan file ...
% End Save floorplan data ... (date=11/07 18:35:23, total cpu=0:00:00.0, real=0:00:01.0, peak res=1908.4M, current mem=1908.4M)
Saving Drc markers ...
... 38 markers are saved ...
... 0 geometry drc markers are saved ...
... 0 antenna drc markers are saved ...
% Begin Save placement data ... (date=11/07 18:35:24, mem=1908.4M)
*** Saving StdCellPlacement Binary (version 2) ...
Save Adaptive View Pruning View Names to Binary file
% End Save placement data ... (date=11/07 18:35:24, total cpu=0:00:00.0, real=0:00:00.0, peak res=1908.8M, current mem=1908.8M)
% Begin Save routing data ... (date=11/07 18:35:24, mem=1908.8M)
Saving route file ...
*** Completed saveRoute (cpu=0:00:00.0 real=0:00:00.0 mem=2036.2M) ***
% End Save routing data ... (date=11/07 18:35:24, total cpu=0:00:00.0, real=0:00:00.0, peak res=1908.9M, current mem=1908.9M)
Saving property file DBS/cts.enc4.dat/register file.prop
*** Completed saveProperty (cpu=0:00:00.0 real=0:00:00.0 mem=2039.2M) ***
% Begin Save power constraints data ... (date=11/07 18:35:24, mem=1910.2M)
% End Save power constraints data ... (date=11/07 18:35:24, total cpu=0:00:00.0, real=0:00:00.0, peak res=1910.2M, current mem=1910.2M)
Generated self-contained design cts.enc4.dat
% End Save design ... (date=11/07 18:35:24, total cpu=0:00:00.2, real=0:00:02.0, peak res=1938.9M, current mem=1912.3M)
*** Message Summary: 0 warning(s), 0 error(s)

Innovus 4> 
```

Applications Places Innovus(TM) Implementation System 21.15 - /home/student/22BEC1204_ASIC_DESIGN/MPD - register_file - ic312...

Innovus(TM) Implementation System 2 NanoRoute -- ic312.vit.ac.in

File View Edit Partition Floorplan Power Place ECO Clock Route Timing Verify PVS Tools Windows Flowgs Help

Layout

Net

Snap

A

Routing Phase

- ☒ Global Route
- ☒ Detail Route End Iteration 1
- ☒ Post Route Optimization ☒ Optimize Via ☒ Optimize Wire

Concurrent Routing Features

- ☒ Fix Antenna
- ☐ Insert Diodes Diode Cell Name
- ☐ Timing Driven Effort 5 Congestion Timing S.M.A.R.T.
- ☐ SI Driven
- ☐ Litho Driven
- ☐ Post Route Litho Repair

Routing Control

- ☐ Selected Nets Only Bottom Layer 1 Top Layer 9
- ☐ ECO Route
- ☐ Area Route Area Select Area and Route

Job Control

- ☒ Auto Stop
- Number of Local CPUs: 1
- Number of CPUs per Remote Machine: 1
- Number of Remote Machines: 0
- Set Multiple CPU...

OK Apply Attribute Mode Save Load Cancel Help

select Instance Enter Instance name

Click to select single object. Shift+Click to deselect multiple objects.

Append coord: 0,0 radius 150 Goto 49.96150, 124.27290 Sel: 0 Timing Analysis

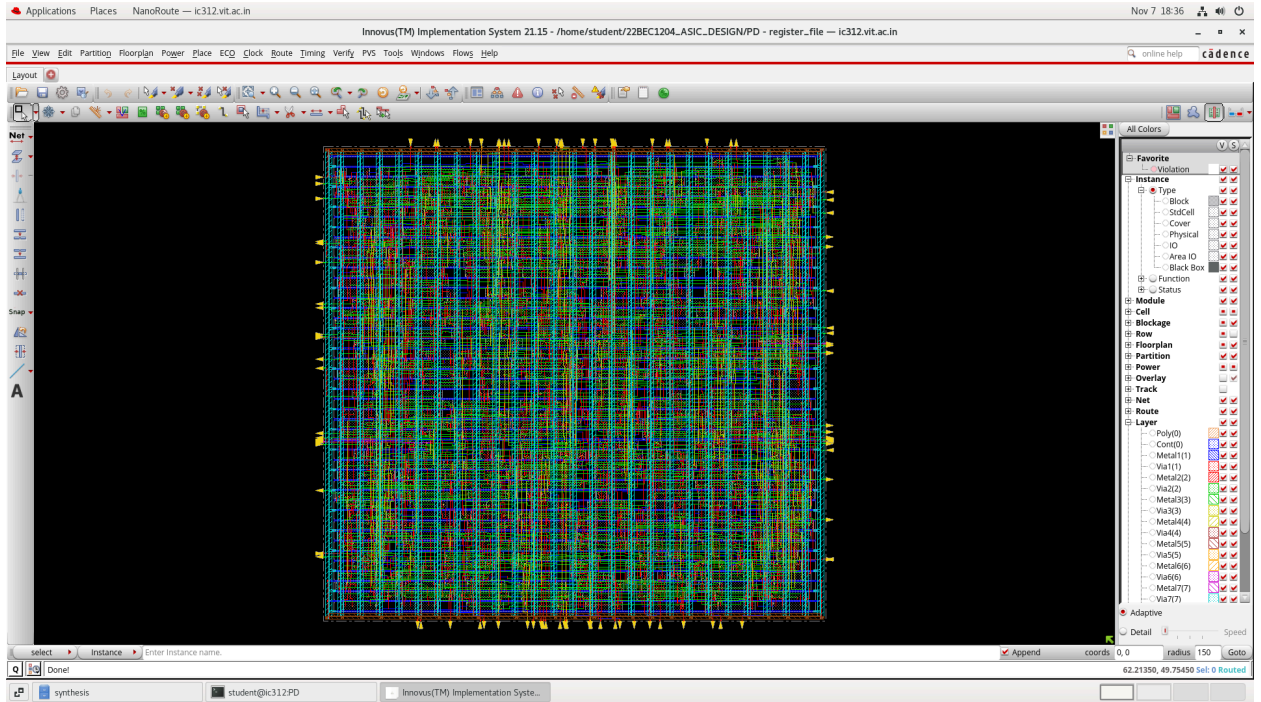
synthesis student@ic312PD Innovus(TM) Implementation System NanoRoute -- ic312.vit.ac.in

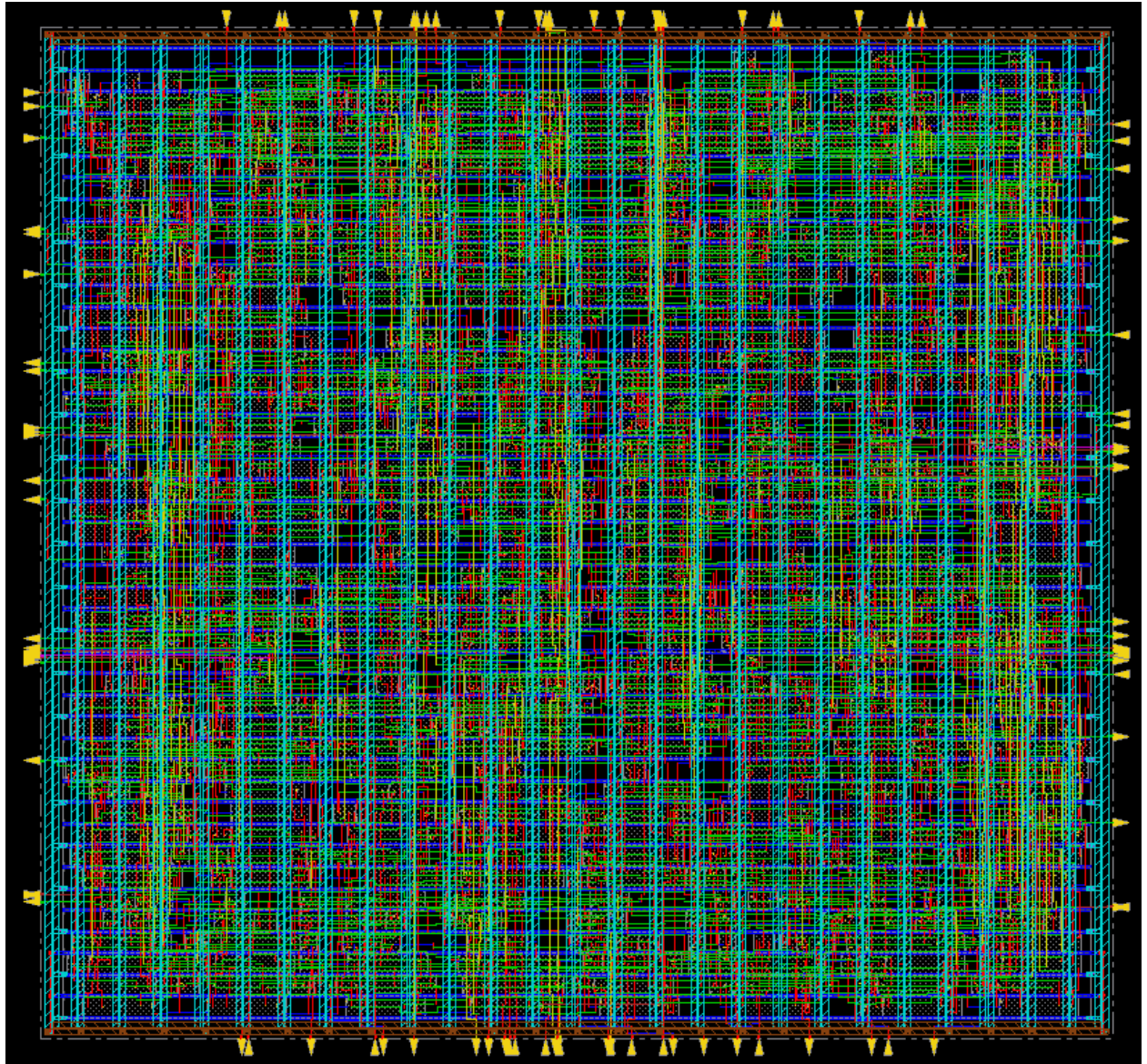
Favorite

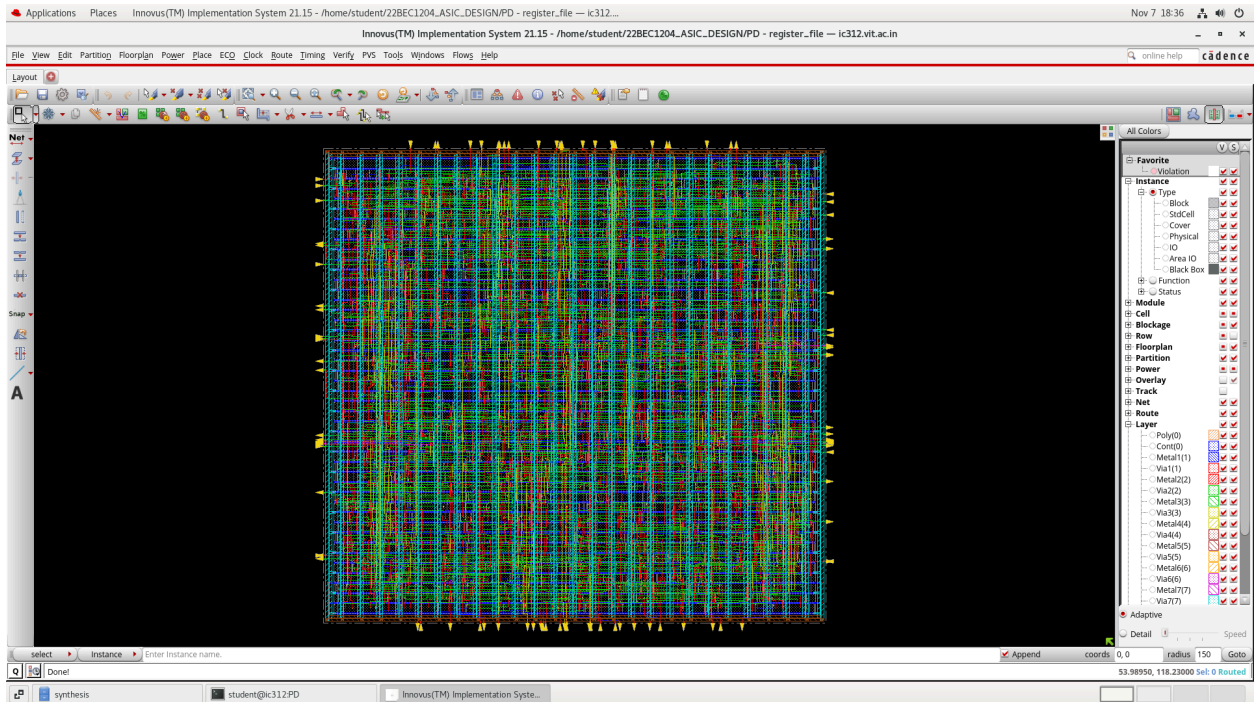
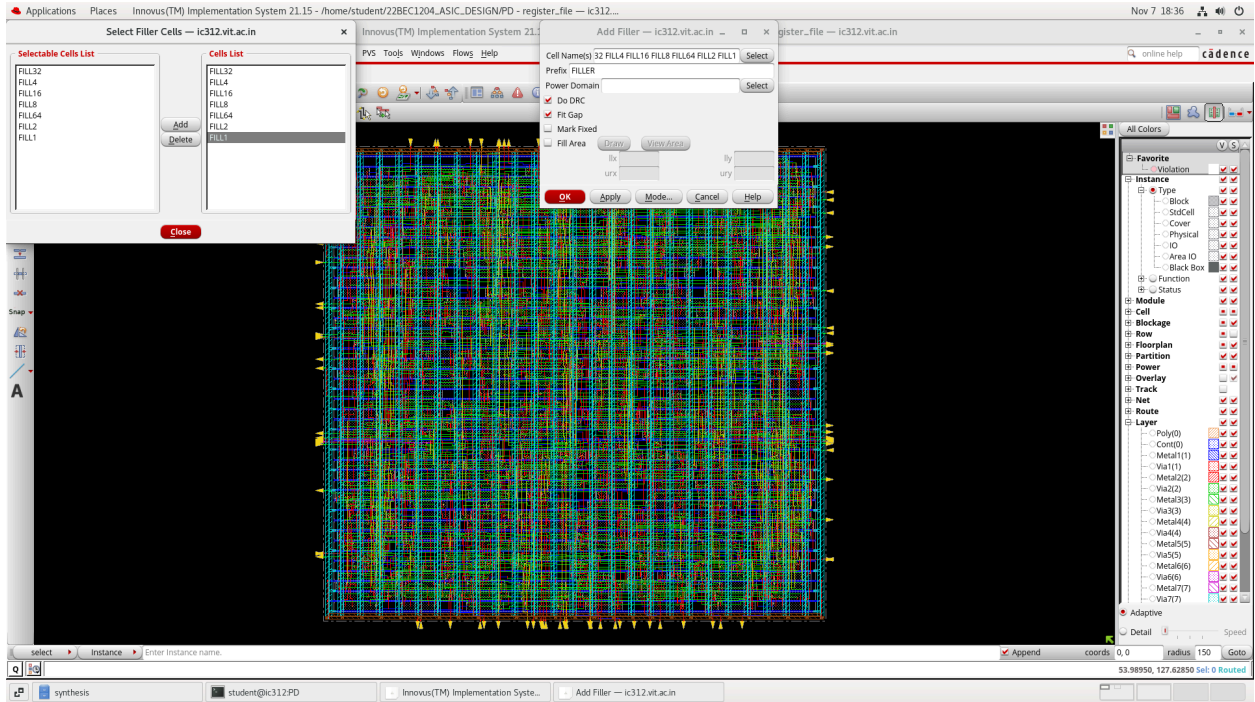
- Violation
- Type
- Instance
- Block
- StdCell
- Cover
- Physical
- IO
- Area IO
- Black Box
- Function
- Status
- Module
- Cell
- Blockage
- Flow
- Floorplan
- Partition
- Power
- Overlay
- Track
- Net
- Route
- Layer
- Poly(8)
- Cont(0)
- Metal(11)
- Via(11)
- Mena(22)
- Via(2)
- Metal(3)
- Via(3)
- Metal(4)
- Via(4)
- Metal(5)
- Via(5)
- Metal(6)
- Via(6)
- Metal(7)
- Via(7)

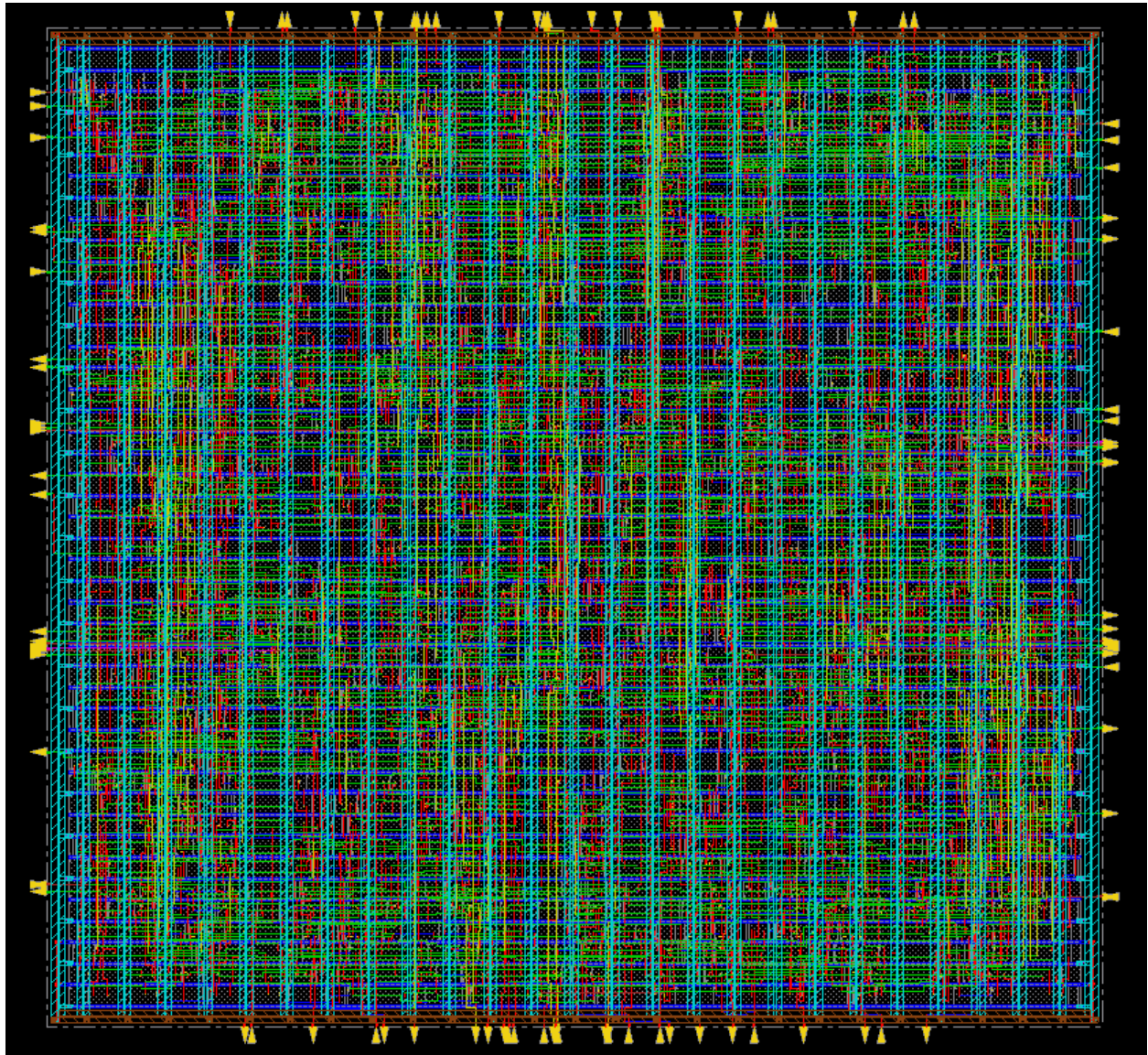
Adaptive

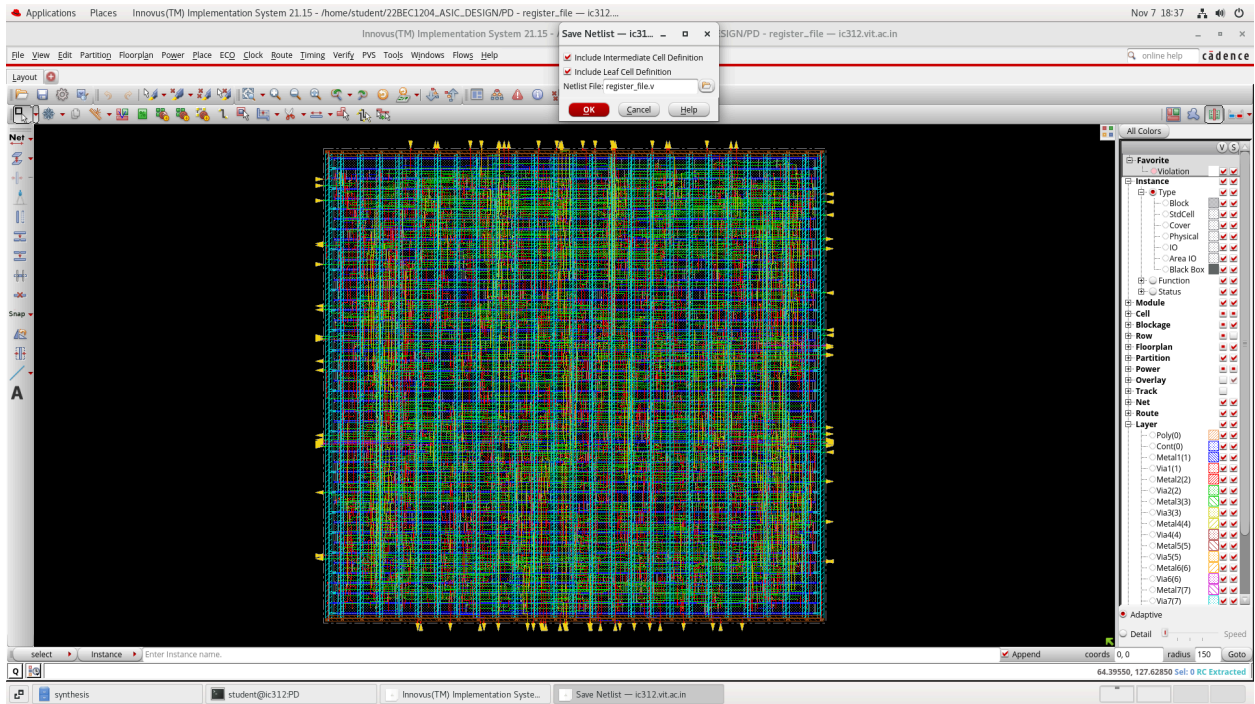
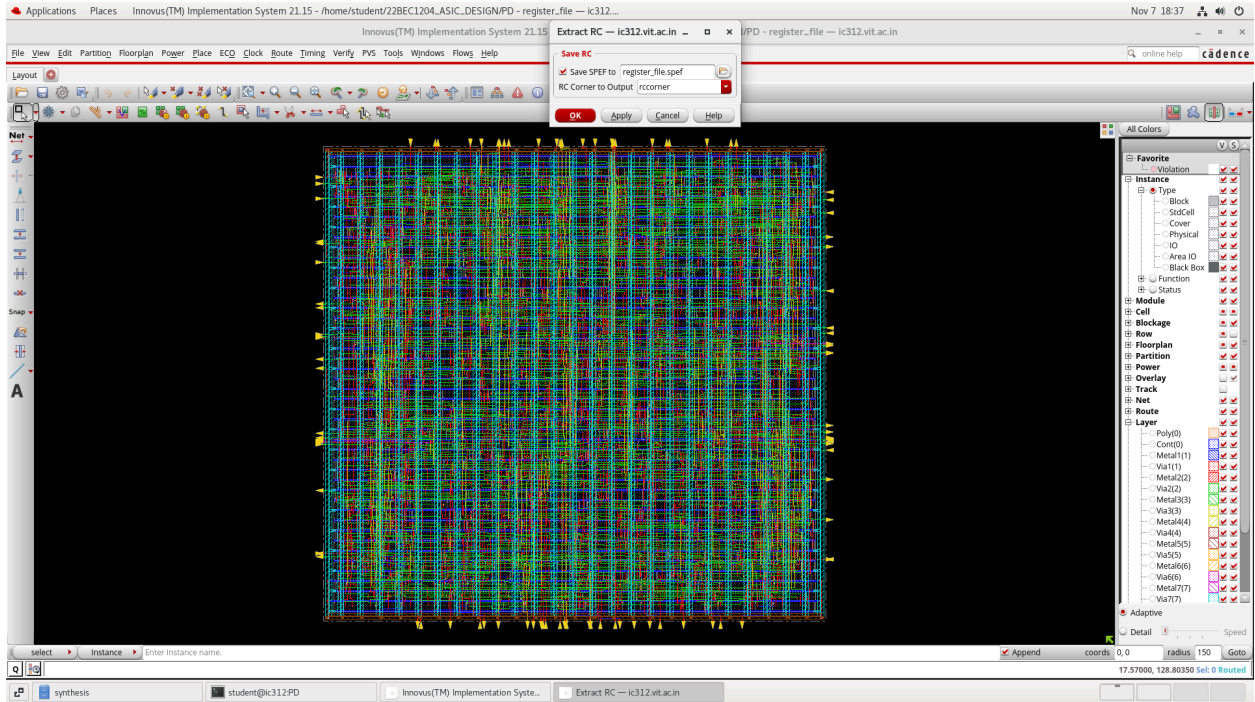
Detail Speed

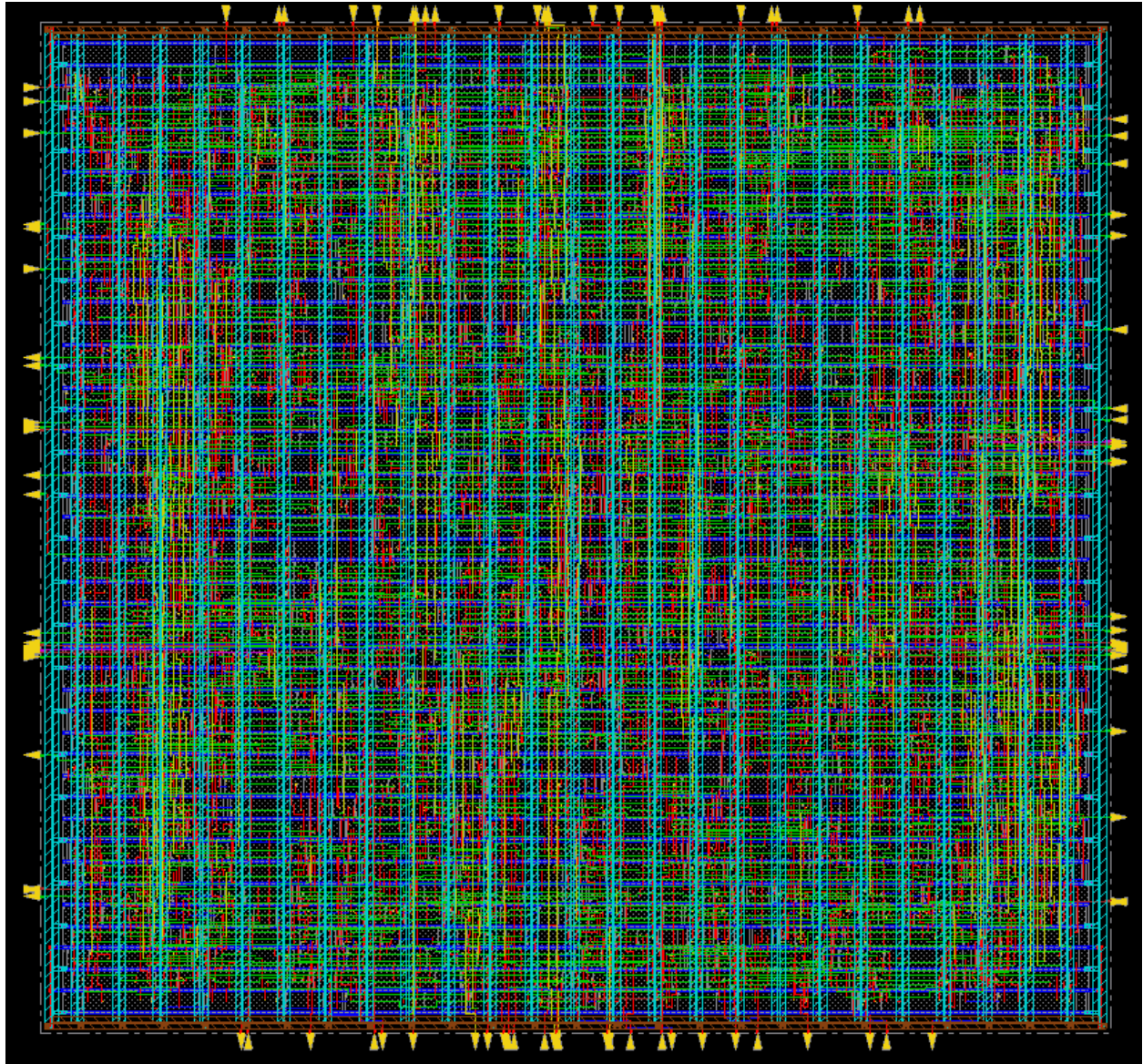


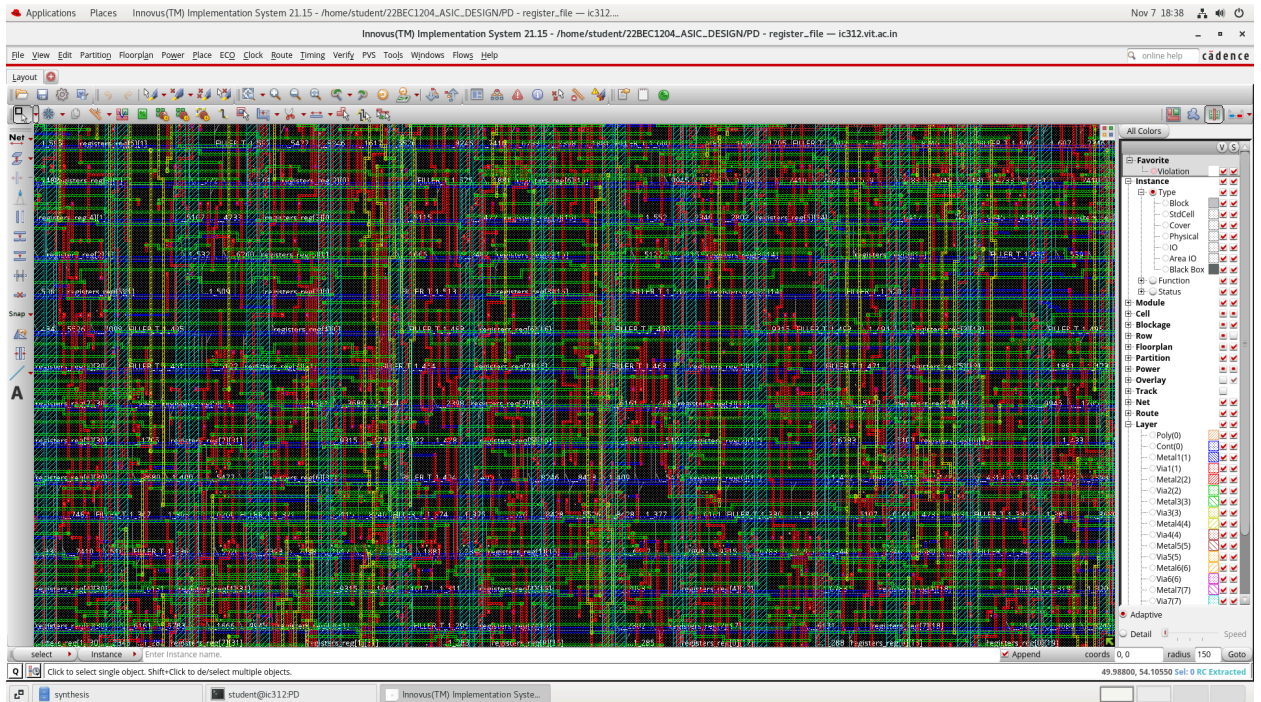
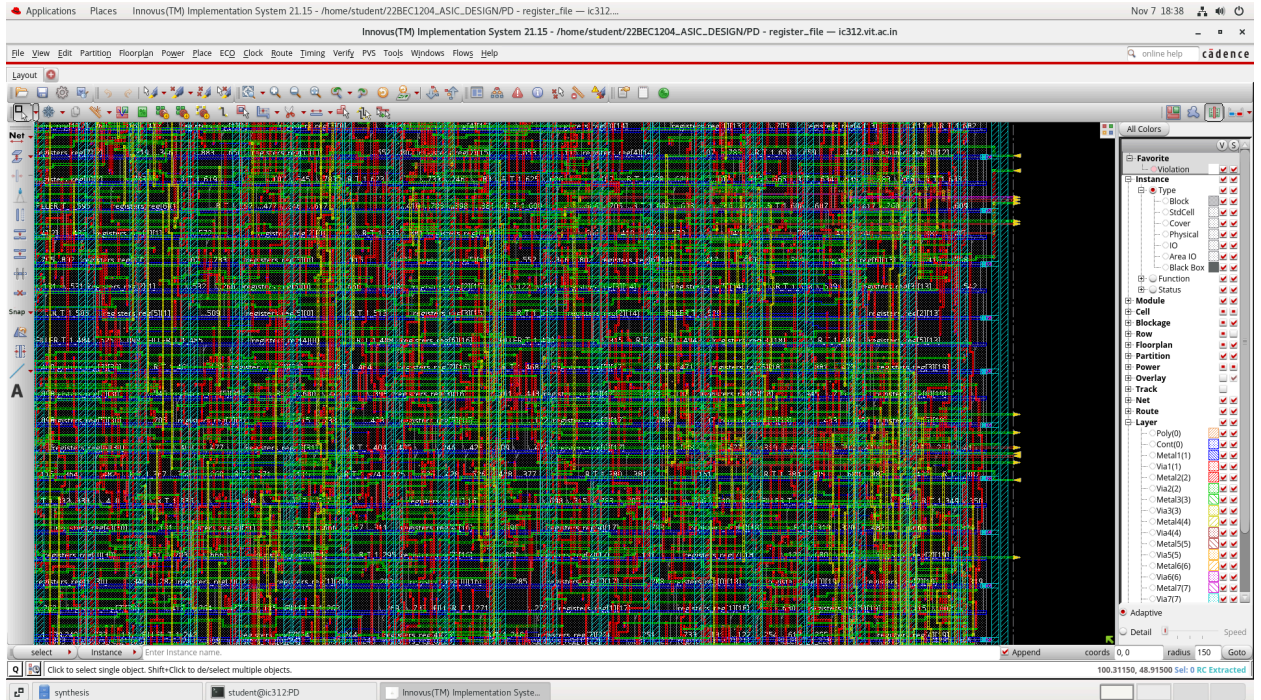


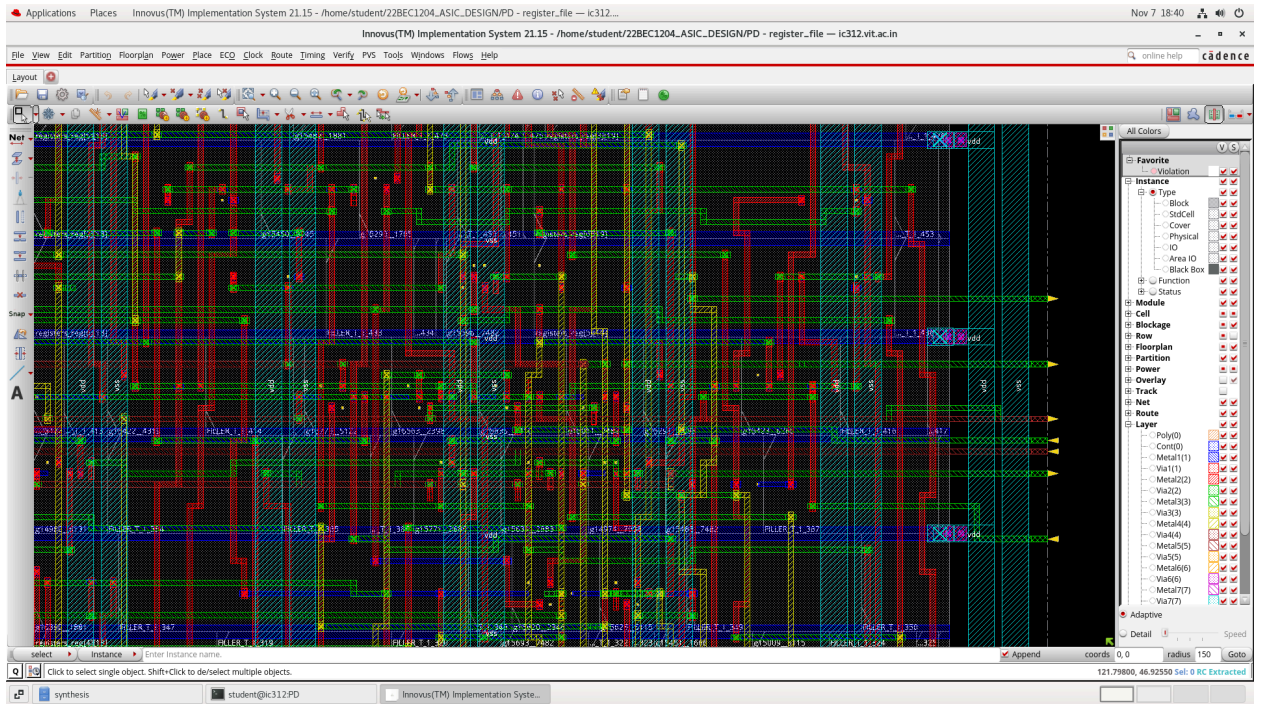
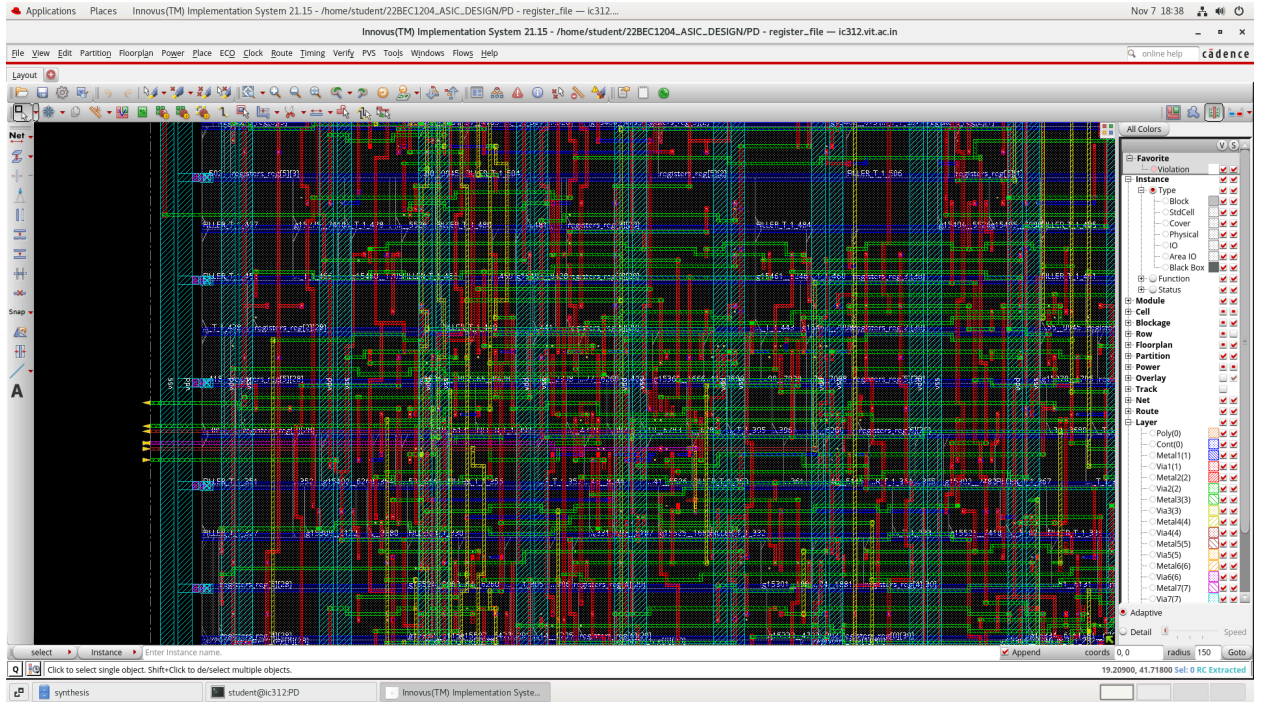


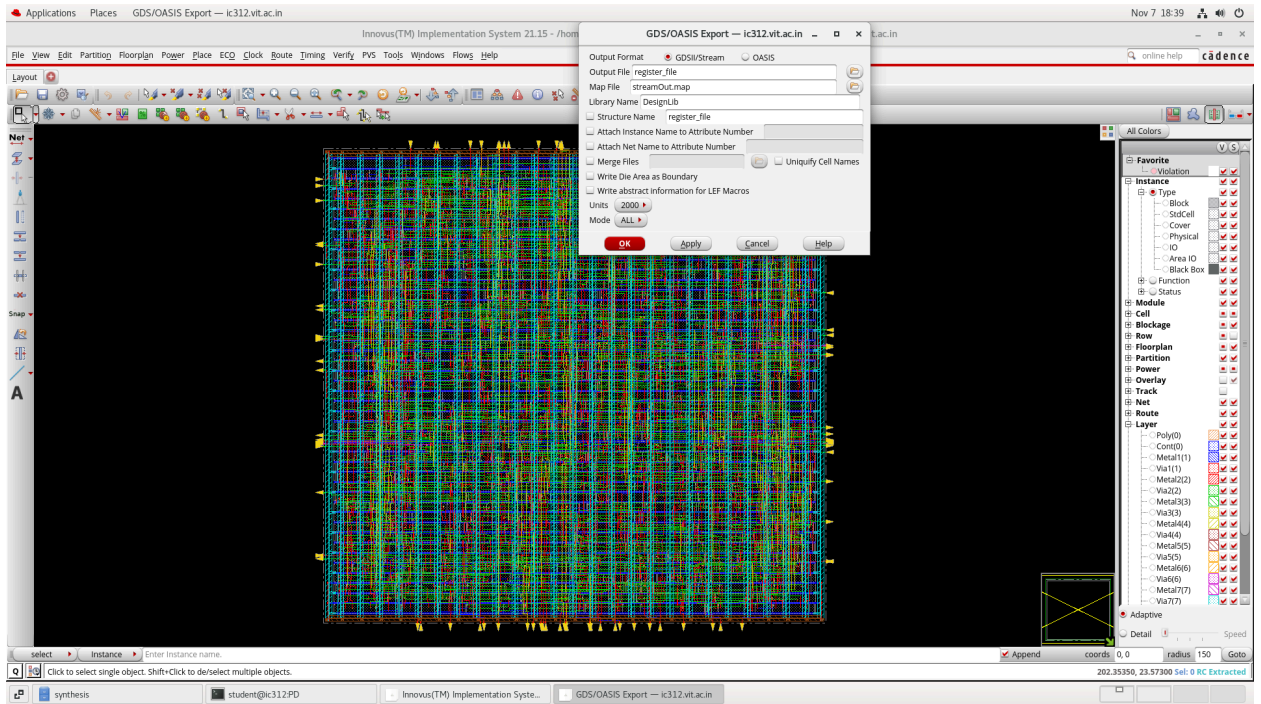
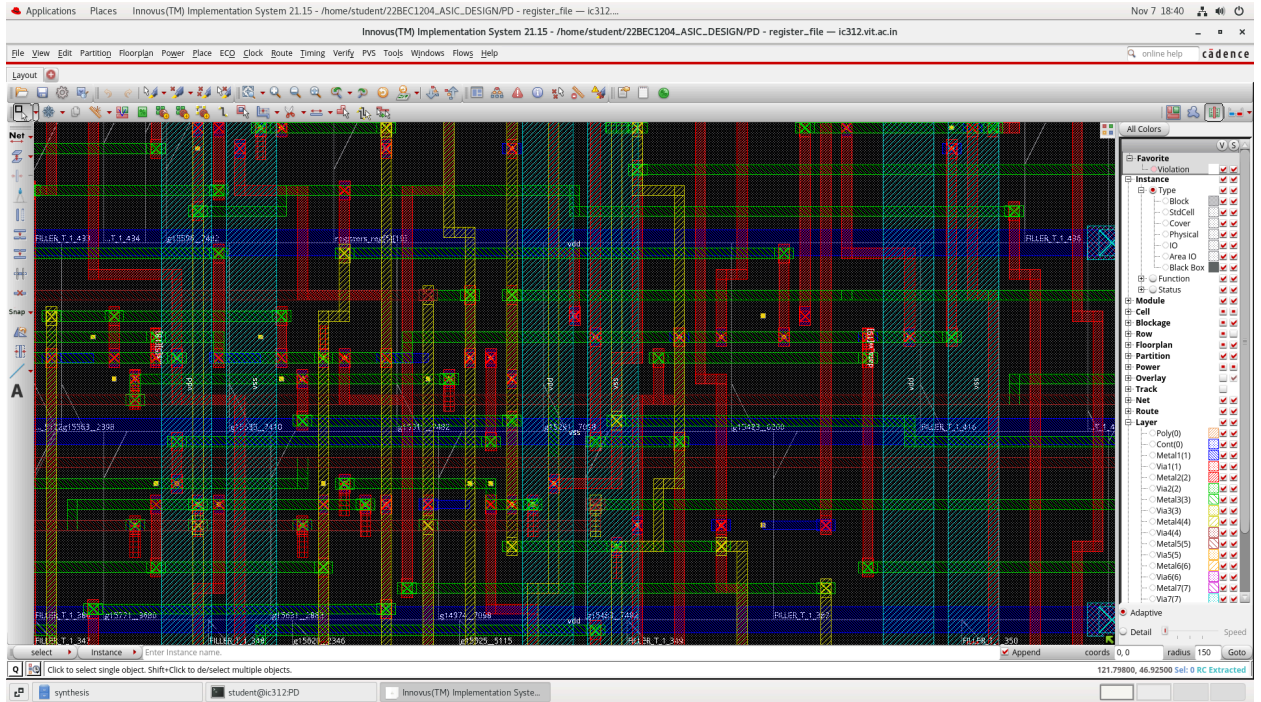












```
Applications Places Terminal Nov 7 18:40 student@ic312:PD
File Edit View Search Terminal Help
metal layer Metal1 604
metal layer Metal2 8478
metal layer Metal3 5199
metal layer Metal4 775
metal layer Metal5 41
metal layer Metal6 12
Via Instances 8749
Special Nets 112
metal layer Metal1 108
metal layer Metal2 12
Via Instances 384
Metal Fills 0
Via Instances 0
Metal FillOPCs 0
Via Instances 0
Metal FillDRCs 0
Via Instances 0
Text 1029
metal layer Metal1 60
metal layer Metal2 635
metal layer Metal3 294
metal layer Metal4 19
metal layer Metal5 15
metal layer Metal6 6
Blockages 0
Custom Text 0
Custom Box 0
Trim Metal 0
***WARN: (IMP0605-1176): There are 4 empty cells. Check innovus.log# for the details.
It is probably because your mapping file does not contain corresponding rules.
Use default mapping file(without option -mapFile) to output all information of a cell.
Type 'man IMP0605-1176' for more detail.
#####Streamout is finished!
innovus > 
```

7.Inference

A synthesizable 8x32-bit 2R/1W register file was successfully designed, verified, and implemented. The design correctly modeled the asynchronous active-low reset and synchronous write logic.

This lab demonstrated the importance of thorough testbenches. The initial testbench produced a "passing" simulation but failed to achieve high code coverage. By enhancing the testbench to cover more corner cases (like reset-override) and bit-toggling patterns, we gained much higher confidence in the design's correctness and achieved better coverage metrics. The design passed the full RTL-to-GDSII flow, resulting in a testable, physically-correct GDSII layout.